41107 Marine and Ocean Engineering
Homework Notes - First Part of the Course

Based on the homework prompts shown in the lecture screenshots from the first part of the course.

These notes are meant as **study notes**, not as a line-by-line reproduction of the textbook. The goal is to explain the logic behind the homework questions, especially the physical meaning behind the plots and formulas. Where the screenshots only show the final formula or the homework prompt, the notes explain the derivation route and the interpretation that matters for the exam.

Contents

1. Steady-current cylinder flow: Strouhal number and vortex shedding frequency

1.1. Definition

The Strouhal number is

$$St = \frac{f_v D}{U},$$

where:

- f_v is the vortex shedding frequency,
- D is the cylinder diameter,
- U is the undisturbed free-stream velocity.

Physically, St measures how fast the wake sheds vortices compared to the time scale of the incoming flow. If the flow speed is fixed and St increases, then vortices are being shed more frequently.

1.2. How St varies with Re

The Reynolds number is

$$Re = \frac{UD}{\nu}.$$

The variation of St with Re follows the change in **boundary layer behaviour**, **separation point**, and **wake width**. This is the central idea behind the homework prompt.

(a) Laminar vortex street / very low shedding Reynolds numbers

At low Re the wake is still organizing itself into a periodic shedding process. The shedding frequency becomes better defined as the classical von Karman street appears. As the wake becomes more regular, St rises from lower values toward about 0.2.

(b) Subcritical regime

In the subcritical regime, St is approximately constant around 0.2. This happens because:

- the boundary layer is still mainly laminar up to separation,
- separation occurs relatively early on both sides of the cylinder,
- the wake structure changes only slowly with increasing Re.

So the characteristic time scale of vortex formation does not change much relative to U/D, and therefore St stays nearly constant.

For the exam, the simplest explanation is: **in the subcritical regime the separation point is nearly fixed, so the wake width and shedding mechanism remain similar, which keeps St almost constant.**

(c) Critical transition

Near the critical Reynolds number, the boundary layer starts to transition to turbulence before it separates. A turbulent boundary layer has more momentum near the wall, so it can resist the adverse pressure gradient longer. Therefore:

- separation is delayed,
- the wake becomes narrower,
- the vortex shedding time scale decreases,
- therefore the shedding frequency increases,
- and thus St jumps upward.

This is why the graph shows a sudden rise in St around the critical region.

(d) Supercritical regime

After the jump, St stays high for a while, then tends to decrease again. The reason is that the flow reorganizes into a new pattern with turbulent boundary layers and a modified separation behaviour. The wake is still narrower than in the subcritical case, but the separation position and wake dynamics are no longer changing in the abrupt way seen during the critical jump.

(e) Transcritical regime

At even higher Re, the wake becomes fully turbulent and more three-dimensional. The value of St tends to move toward a more moderate level again. The detailed scatter in experiments is normal because the wake is now very sensitive to roughness, turbulence, and small asymmetries.

1.3. Why do we get this overall pattern?

Because **the shedding frequency is controlled by the wake formation process**, and the wake formation process is controlled by:

- when the boundary layer separates,
- whether the boundary layer is laminar or turbulent,
- how wide the wake is,
- how coherent the vortex street is.

So the St-Re curve is really a map of how the separation and wake physics change with Reynolds number.

2. Effect of surface roughness on flow regimes and on St

Surface roughness, often described by k_s or relative roughness k_s/D, changes the flow because it triggers earlier transition to turbulence in the boundary layer.

2.1. Main physical effect

Compared with a smooth cylinder, a rough cylinder causes:

- earlier boundary layer transition,
- earlier delay of separation,

- therefore earlier onset of critical and supercritical behaviour,
- and consequently the changes in St occur at lower Re.

2.2. How to explain the roughness plot

The right way to read the roughness plot is:

1. First understand the **smooth** curve using the explanation above.
2. Then note that increasing roughness shifts the boundary-layer transition forward.
3. That means the "critical" behaviour happens earlier, i.e. at lower Reynolds number.
4. So the jumps or plateaus in St are shifted left when k_s/D increases.

Roughness does not create a completely new physics. It mainly changes **when** the boundary layer becomes turbulent, and therefore it shifts the regime boundaries in Re.

3. Drag coefficient of a smooth cylinder and drag crisis

3.1. Why C_D is roughly constant in the subcritical regime

The drag coefficient is

$$C_D = \frac{F_D}{\frac{1}{2}\rho U^2 D},$$

for force per unit length on the cylinder.

In the subcritical regime, C_D is roughly constant, typically around 1. The reason is:

- the boundary layer is laminar up to separation,
- separation occurs early,
- the wake remains broad,
- pressure drag dominates,
- and the overall pressure distribution changes only modestly with Re.

So the normalized drag, C_D, does not change much.

3.2. Why drag crisis occurs

The **drag crisis** is the sudden drop in C_D near the critical Reynolds number. It occurs because:

1. the boundary layer transitions to turbulence before separation,
2. a turbulent boundary layer has more near-wall momentum,
3. so separation is delayed,
4. the wake becomes narrower,
5. the base pressure behind the cylinder increases,

6. the pressure difference between front and back is reduced,
7. and therefore the pressure drag drops sharply.

This is one of the most important physical explanations in the first part of the course.

3.3. Upper regimes after drag crisis

After the crisis, C_D becomes much smaller than in the subcritical regime. Then, at still higher Reynolds numbers, the wake becomes more complicated and C_D may increase slightly again. This later rise is associated with a new balance between turbulent separation, wake structure, and skin-friction effects.

If asked "Why is C_D almost constant in subcritical flow?" answer: **because separation occurs early and at nearly the same angular position over a wide range of Re, so the wake width and pressure drag remain similar.**

4. Oscillatory flow, KC number, and flow regimes before vortex shedding

4.1. Definition and physical meaning of KC

The Keulegan-Carpenter number is

$$KC = \frac{U_m T_w}{D} = \frac{2\pi a}{D},$$

where:

- U_m is the maximum oscillatory velocity,
- T_w is the oscillation period,
- a is the horizontal particle-motion amplitude,
- D is the cylinder diameter.

So KC is essentially a **stroke-to-diameter ratio**. It tells you how far the water moves in one half/full cycle relative to cylinder size.

4.2. Why KC matters physically

- If KC is very small, particles only move a short distance relative to the cylinder diameter. The flow may remain attached and can even show no separation.
- If KC increases, the oscillatory excursion becomes large enough for separation to occur.
- If KC becomes still larger, vortices can grow, detach, and be shed.

So in oscillatory flow, KC plays a role similar to how travel distance or stroke length controls the wake pattern.

4.3. Flow regimes before vortex shedding

For a given Re, the oscillatory-flow regimes prior to full shedding can be summarized conceptually as follows:

- **Very small KC:** no separation, creeping or attached oscillatory flow.
- **Moderate KC:** symmetric vortices can form behind the cylinder but may remain attached.
- **Higher KC before shedding:** asymmetric vortices and stronger separation appear.

The important idea is that the wake becomes progressively more nonlinear as the oscillatory excursion increases.

5. Vortex shedding regimes in oscillatory flow

5.1. When shedding starts

A standard rule from the lecture material is:

$$KC > 7 \quad \Rightarrow \quad \text{vortex shedding occurs.}$$

5.2. Single-pair, double-pair, and higher regimes

Once shedding starts, the number of vortex pairs shed in one oscillation period can change with KC. Typical labels are:

- single-pair regime,
- double-pair regime,
- three-pair regime,
- and so on.

The meaning is literal: how many vortex pairs are released during one full wave cycle.

5.3. Why the regime changes with KC

As KC increases, the fluid has more time and more excursion distance during each half-cycle. That allows more vortical structures to grow and separate before the flow reverses. Hence:

- low shedding KC: only one pair fits into one cycle,
- larger KC: two pairs can form,
- even larger KC: three or more pairs can appear.

5.4. Single-pair regime: how to explain it

In the single-pair regime, one pair of vortices is effectively shed during one full oscillation period. During the flow reversal, recently shed vortices can be swept back toward the cylinder, which strongly affects the lift force. This is why the force record may show more structure than a simple "one event per period" picture suggests.

6. Normalized lift frequency N_L

6.1. Definition

The normalized lift frequency is

$$N_L = \frac{f_L}{f_w},$$

where:

- f_L is the dominant frequency of the lift force,
- f_w is the wave frequency.

6.2. Why N_L equals the number of lift oscillations per wave cycle

One wave cycle lasts

$$T_w = \frac{1}{f_w}.$$

The lift oscillation period is

$$T_L = \frac{1}{f_L}.$$

So the number of lift oscillations occurring during one wave period is

$$\frac{T_w}{T_L} = \frac{1/f_w}{1/f_L} = \frac{f_L}{f_w} = N_L.$$

Therefore, N_L is exactly the number of oscillations in lift per wave cycle.

6.3. Why a single-pair regime can still give $N_L = 2$

This is a point many students find confusing. A single-pair regime means one vortex pair is shed in one wave period. But the lift force can still oscillate twice in that period because:

- one contribution comes from the shedding process itself,
- another strong contribution comes after flow reversal when the most recently shed vortices interact again with the cylinder.

So the number of **lift-force oscillations** is not always the same thing as the number of **net vortex pairs shed**.

6.4. Range of f_L from the lecture estimate

If

$$f_w \in [0.04, 0.2] \text{ Hz}, \qquad N_L \in [2, 20],$$

then

$$f_L = N_L f_w.$$

Hence the approximate range is

$$f_L \in [2 \cdot 0.04, 20 \cdot 0.2] = [0.08, 4] \text{ Hz}.$$

7. Hydrodynamic mass and Morison-type force balance

7.1. Potential-flow result for a circular cylinder

For very small KC (no separation, potential-like flow), the hydrodynamic mass per unit length of a circular cylinder is

$$m' = \rho \pi r_0^2 = \rho A,$$

where A is the cross-sectional area. If one writes

$$m' = \rho C_m A,$$

then the potential-flow result gives

$$C_m = 1.$$

7.2. Important notation warning

Different books use different symbols:

- some use C_m for **added mass coefficient**,
- some use C_M for the **total inertia coefficient** in Morison's equation,
- often they are related by
$$C_M = 1 + C_m.$$

This explains why some plots show a "theoretical" asymptotic value near 2 instead of 1. The extra 1 comes from the Froude-Krylov contribution.

7.3. Inertia-dominated versus drag-dominated regime

A useful estimate from the slides is

$$\frac{\max(F_M)}{\max(F_D)} = \frac{\pi^2}{KC}\frac{C_M}{C_D} \approx \frac{20}{KC} \quad \text{for small } KC.$$

So:

- if $KC \ll$ 20–30, inertia dominates,
- if $KC \gg$ 20–30, drag dominates.

Physical explanation

For small KC, the excursion is short but the acceleration is large, so the fluid force is mainly associated with accelerating fluid around the body. There is little time/distance for a strong separated wake to develop, so drag is weaker.

For large KC, the excursion is long, separation is strong, and the wake has time to develop. Then viscous drag and wake effects become dominant.

8. Variation of C_D and C_M with KC

8.1. Behaviour of C_D

A typical interpretation of the C_D-KC curve is:

- **Very small KC:** no separation, drag is relatively small.
- **Around onset of separation:** C_D changes rapidly because the wake starts to form.
- **Around vortex shedding onset and established shedding:** C_D can show a hump because the separated wake becomes strong and organized.
- **Large KC:** C_D tends toward a plateau closer to a steady-current-like value.

8.2. Why the hump appears

The hump is associated with the most energetic range of separated wake formation. As coherent transverse vortex shedding becomes established, pressure drag increases. Once the wake is fully developed and the oscillatory cycle becomes more like repeated separated-flow events, the coefficient approaches a large-KC limit.

8.3. Behaviour of C_M and negative added mass

The total inertia coefficient C_M is close to its potential-flow value at small KC. Around intermediate KC (often around the vortex-shedding range) C_M can drop strongly. If one defines added mass coefficient as

$$C_m = C_M - 1,$$

it can even become negative.

Why can added mass appear negative?

Because once vortex shedding is strong, the hydrodynamic force is no longer just the clean potential-flow acceleration force. The vortices generate pressure fields that are out of phase with the acceleration. If the vortex-induced contribution opposes the classical inertia contribution strongly enough, the fitted or inferred added-mass coefficient becomes negative.

Negative added mass does **not** mean the cylinder has negative physical mass. It means the **fluid-force component extracted from the data is out of phase in such a way that it reduces the effective inertia term**. This is a wake-effect phenomenon.

9. Correlation length

9.1. Correlation length in steady current as function of Re

Correlation length measures how far along the span the fluctuating force remains coherent. Large correlation length means the wake is organized similarly over a long span. Small correlation length means the wake varies quickly along the span.

The trend with Re can be explained by wake coherence:

- near onset of organized vortex streets, coherence is large,
- in turbulent and three-dimensional wakes, coherence decreases,
- near critical/high-Re regimes, separation and turbulence destroy spanwise organization, so correlation length becomes small.

9.2. Correlation length in oscillatory flow

The oscillatory-flow plots show that the correlation length is largest near the middle of the vortex-shedding-regime intervals (for example around single-pair, double-pair, etc.). This happens because the shedding pattern is then most repeatable and stable from cycle to cycle.

Near transitions between regimes, the pattern is less stable, more irregular, and the correlation decreases.

High correlation length = highly repeatable shedding pattern. This is the main message behind the oscillatory-flow correlation plots.

10. Pressure on the cylinder surface in waves and the phase shift δ

10.1. Why $p = -\rho\partial\phi/\partial t$ in linear theory

In linear potential wave theory, Bernoulli's equation is linearized by neglecting the quadratic velocity term. That gives the dynamic pressure as

$$p = -\rho\frac{\partial\phi}{\partial t}.$$

This is valid when wave amplitude is small enough that the $|\nabla\phi|^2$ term is of higher order and can be neglected.

10.2. What the phase shift δ means in the total-force formula

The total force on the unit height of the cylinder is written in the form

$$F_x = (\text{amplitude})\cos(\omega t - \delta).$$

Here δ is the **phase shift** between the force and the incident harmonic wave motion.

Physically, this means the force does not reach its maximum exactly when the undisturbed wave quantity (such as elevation, velocity, or acceleration) does. The cylinder disturbs the flow, and this diffraction effect changes both:

- the **magnitude** of the force,
- and the **timing** of the force.

So:

- amplitude factor $A(kr_0)$ tells you how much the force is amplified or attenuated,
- phase shift $\delta(kr_0)$ tells you how much the force is delayed or advanced.

If asked "What does δ mean?" answer: **it is the phase lag (or lead) of the total force relative to the incident wave forcing, caused by diffraction around the cylinder.**

11. Vibration equation, damping, and reduced velocity

11.1. Basic vibration equation

The standard single-degree-of-freedom vibration equation is

$$m\ddot{y} + c\dot{y} + ky = F(t),$$

where:

- m is mass,
- c is viscous damping coefficient,
- k is spring stiffness,
- y is displacement,
- $F(t)$ is external forcing.

This is the reference model behind much of the vibration interpretation in the course.

11.2. Structural damping and why ζ is related to dissipated energy

The damping ratio ζ is a dimensionless measure of damping intensity. For a lightly damped system, one can show that

$$\zeta \propto \frac{E_{\text{dissipated per cycle}}}{E_{\text{stored}}}.$$

More precisely, for linear light damping,

$$\zeta = \frac{1}{4\pi}\frac{\Delta E}{E},$$

where ΔE is the energy lost in one cycle and E is the maximum stored energy.

So the bigger ζ is, the larger the fraction of energy removed each cycle. That is why heavily damped systems decay quickly.

11.3. Reduced velocity and lock-in

A common reduced velocity is

$$V_r = \frac{U}{Df_n},$$

where f_n is the natural frequency.

Lock-in occurs when the vortex-shedding frequency becomes close to the structural natural frequency, so the wake and the structure synchronize. Then the oscillation amplitude becomes large.

In water, the response differs from the air case because:

- added mass changes the effective natural frequency,
- fluid damping is important,
- the cylinder motion changes the relative velocity,
- therefore the wake and the structure interact more strongly.

12. Cross-flow vibrations at fixed KC (example: $KC = 20$)

12.1. Why the frequency response looks step-like or zig-zag

The key definitions are

$$N = \frac{f}{f_w}, \qquad V_r = \frac{U_m}{f_n D}, \qquad KC = \frac{U_m}{f_w D}.$$

If the system is in lock-in, then $f \approx f_n$, so

$$N = \frac{f}{f_w} \approx \frac{f_n}{f_w} = \frac{U_m/(V_r D)}{U_m/(KC\, D)} = \frac{KC}{V_r}.$$

Therefore, at fixed KC, increasing V_r makes N decrease.

But the wake prefers stable states with nearly integer numbers of oscillations per wave cycle, such as $N = 4$, $N = 3$, $N = 2$. So instead of following one smooth line, the response stays near one branch for a while, then jumps to the next. This creates the zig-zag or step-like pattern.

12.2. Why the amplitude has multiple peaks

Each branch ($N = 4$, $N = 3$, $N = 2$) has its own lock-in window where the excitation is effective. So the amplitude curve shows several peaks, each one associated with a different synchronized response mode. The drops between peaks occur when the system leaves one lock-in branch before entering the next.

12.3. Why the response looks "strange" compared with steady current

In steady current, there is one main flow time scale and the classical lock-in picture is more continuous. In waves, there is an additional periodic forcing time scale f_w. Because the wake must organize itself relative to this oscillatory cycle, the response is quantized into branches labelled by N. That is why the wave case looks more "step-like" than the steady-current case.

12.4. How to reconcile with steady current ($KC \to \infty$)

As $KC \to \infty$, the oscillatory stroke becomes extremely large and the flow tends toward a steady-current-like situation. Then the explicit counting of oscillations per wave cycle becomes less meaningful, and the response tends toward the standard steady-current VIV picture.

13. In-line vibrations

The lecture distinguishes between:

- **in-line vibrations**, which are the higher-frequency oscillations in the streamwise direction caused by the fluctuating force system,
- **in-line motion**, which may follow the lower-frequency oscillatory flow itself.

The important idea is that cross-flow and in-line responses are not the same phenomenon. Cross-flow response is usually associated with lift and is often dominant in VIV, whereas in-line response is linked to the in-line force component and may contain higher-frequency content.

14. Why an iterative solution of the equation of motion is needed

The lecture procedure is:

1. Start with KC and Re based on the absolute flow velocity U.
2. Select force coefficients from the usual diagrams.
3. Solve the equation of motion and get displacement $x(t)$.
4. Compute the relative velocity $U - \dot{x}$.
5. Update KC and Re using that relative velocity.
6. Select new coefficients and iterate until convergence.

14.1. Why this is necessary

Because the hydrodynamic force depends on the **relative motion between fluid and structure**. But the structure motion depends on the force. So the problem is coupled:

$$\text{force} \rightarrow \text{motion} \rightarrow \text{relative velocity} \rightarrow \text{new force.}$$

A single calculation is therefore not enough.

The iterative procedure is needed because the coefficients are not constants fixed once and for all. They depend on the relative flow seen by the moving cylinder.

15. Fast exam summary

- $St = f_v D/U$; it is almost constant in the subcritical regime because the separation point is nearly fixed.
- The critical jump in St and the drag crisis both come from transition to a turbulent boundary layer before separation.
- Roughness shifts critical behaviour to lower Reynolds numbers because it trips the boundary layer earlier.
- $KC = U_m T_w/D = 2\pi a/D$ is a stroke-to-diameter ratio.
- Small KC: inertia-dominated, potential-like flow. Large KC: drag-dominated, strong separated wake.
- Vortex shedding starts roughly for $KC > 7$ and the number of vortex pairs shed per cycle increases with KC.
- $N_L = f_L/f_w$ is the number of lift oscillations per wave cycle.
- In oscillatory flow, correlation length is largest in the middle of stable shedding-regime intervals because the pattern is most repeatable there.
- In linear wave diffraction, δ is the phase lag between total force and incident wave motion.
- For light damping, ζ is proportional to the energy dissipated per cycle divided by stored energy.

- At fixed KC, the cross-flow response branches follow integer-like $N = f/f_w$, giving the zig-zag frequency response and multi-peak amplitude response.
- Iteration is needed because the coefficients depend on relative velocity, which depends on the motion itself.

41107 Marine and Ocean Engineering
Homework Notes - Second Part of the Course

Based on the second batch of homework screenshots on floating structures, resistance, wave statistics, oscillations, spectral analysis, and ship strength.

These notes are meant as a **study guide and worked-solution set**. I explain the logic of each homework prompt, solve the examples that can be solved from the screenshots, and clearly mark the places where the screenshot does not contain enough information to reconstruct the full exercise exactly.

Contents

1. Hydrostatic coefficients and waterplane geometry

1.1. Main geometric coefficients of a ship

Three very common hydrostatic form coefficients are:

$$C_B = \frac{\nabla}{LBT}, \qquad C_M = \frac{A_M}{BT}, \qquad C_P = \frac{\nabla}{A_M L} = \frac{C_B}{C_M}.$$

Here:

- ∇ is the displacement volume,
- L is the reference length,
- B is the beam,
- T is the draught,
- A_M is the immersed midship area.

Always remember the physical meaning:

- C_B tells you how "full" the whole underwater block is,
- C_M tells you how "full" the midship section is,
- C_P tells you how the volume is distributed longitudinally once the midship fullness is fixed.

1.2. Worked example: block, prismatic, and midship coefficients

Screenshot statement: a ship in saltwater has

- $L = 150.000\,\text{m}$,
- $B = 22.000\,\text{m}$,
- $T = 9.000\,\text{m}$,

- displacement mass = 24 000.000 t,
- immersed midship area $A_M = 180.000\,\text{m}^2$.

Find C_B, C_P, and C_M.

In saltwater we take $\rho \approx 1.025\,\text{t/m}^3$, so the displacement volume is

$$\nabla = \frac{24000}{1.025} = 23\,414.630\,\text{m}^3.$$

Then

$$C_B = \frac{23414.63}{150 \cdot 22 \cdot 9} = 0.788.$$

Also,

$$C_M = \frac{180}{22 \cdot 9} = 0.909.$$

Finally,

$$C_P = \frac{23414.63}{180 \cdot 150} = 0.867.$$

So the answers are:

$$\boxed{C_B \approx 0.788, \qquad C_M \approx 0.909, \qquad C_P \approx 0.867.}$$

1.3. Waterplane area from tabulated half-breadths

The screenshot with the MATLAB example gives tabulated values of $y(x)$, where $B(x) = 2y(x)$ is the local waterline breadth. The waterplane area is therefore

$$A_{WP} = \int B(x)\,dx = 2\int y(x)\,dx.$$

If the data are tabulated, then numerically you use trapezoidal integration or Simpson's rule.

If x is measured from the aft end of the overall waterline, then the longitudinal centre of area is

$$\bar{x}_{WP} = \frac{\int xB(x)\,dx}{\int B(x)\,dx} = \frac{2\int xy(x)\,dx}{A_{WP}}.$$

Using the values visible in the screenshot and applying trapezoidal integration gives approximately

$$A_{WP} \approx 5246.640\,\text{m}^2, \qquad \bar{x}_{WP} \approx 73.900\,\text{m}$$

measured from the aft end of the full waterline table. Since the aft perpendicular is about 21.600 m from that aft end in the screenshot, the centre is about

$$\bar{x}_{WP} \approx 52.300\,\text{m}$$

forward of AP.

Because the screenshot only shows the table and not the exact numerical method used in the MATLAB file, a slightly different value can appear if the teacher's script uses a different quadrature convention. The method above is still the correct one to study.

2. Floatation, stability, and restoring moment

2.1. Why a ship floats

A ship floats because vertical hydrostatic pressure integrates to an upward resultant buoyancy force F_B, and this balances the ship weight W in equilibrium:

$$F_B = W.$$

The weight acts through the centre of gravity G, and the buoyancy force acts through the centre of buoyancy B.

In upright equilibrium these two forces are equal and opposite on the same vertical line. Once the ship heels, the submerged volume changes shape, so the centre of buoyancy moves. That creates a moment which can be restoring or overturning.

2.2. Important distances and points

The standard vertical distances are:

$$KB, \qquad BM_T, \qquad KM_T, \qquad KG, \qquad GM_T.$$

They are related by

$$KM_T = KB + BM_T, \qquad GM_T = KM_T - KG.$$

For transverse stability,

$$BM_T = \frac{I_x}{\nabla},$$

where I_x is the second moment of area of the waterplane about the ship centreline and ∇ is the displacement volume.

2.3. Why a ship with a higher centre of gravity does not necessarily capsize

A higher G reduces GM, so the ship is *less stable*. But "less stable" does not automatically mean unstable.

The key criterion is the sign of GM for small angles:

- if $GM > 0$, the ship has a restoring moment for small heel and is initially stable,
- if $GM = 0$, it is neutrally stable,
- if $GM < 0$, it is initially unstable.

So a higher centre of gravity may still leave $GM > 0$. In that case the ship heels more easily, but still tends to come back.

2.4. Deriving the metacentric radius

From the wedge argument in the screenshot,

$$\delta M_x^h = \rho g I_x \sin(\delta\varphi).$$

Since the same restoring moment can also be written as

$$\delta M_x^h = \rho g \nabla BM_T \sin(\delta\varphi),$$

we obtain

$$\boxed{BM_T = \frac{I_x}{\nabla}.}$$

This is one of the most important hydrostatics formulas in the course.

2.5. Small-angle restoring lever

For small heel angles,

$$GZ \approx GM_T \sin\varphi.$$

Hence the restoring moment is

$$M_R \approx W\, GM_T \sin\varphi = \rho g \nabla GM_T \sin\varphi.$$

For a wall-sided ship, a more accurate finite-angle correction is

$$GZ = \sin\varphi \left(GM_T + \frac{1}{2} BM_T \tan^2\varphi \right).$$

This is often called the **wall-sided formula**.

2.6. Rectangular barge: expressions for KB, BM_T, and KM_T

For a rectangular box barge of length L, beam B, and draught T:

$$KB = \frac{T}{2}.$$

The waterplane is a rectangle, so

$$I_x = \frac{LB^3}{12}.$$

The displacement volume is

$$\nabla = LBT.$$

Therefore,

$$BM_T = \frac{I_x}{\nabla} = \frac{LB^3/12}{LBT} = \frac{B^2}{12T}.$$

So

$$KM_T = KB + BM_T = \frac{T}{2} + \frac{B^2}{12T}.$$

If the metacentre lies in the waterplane, then $KM_T = T$. Therefore,

$$\frac{T}{2} + \frac{B^2}{12T} = T.$$

This gives

$$\frac{B^2}{12T} = \frac{T}{2} \quad \Rightarrow \quad B^2 = 6T^2 \quad \Rightarrow \quad T = \frac{B}{\sqrt{6}}.$$

With $B = 9.000\,\mathrm{m}$:

$$T = \frac{9}{\sqrt{6}} = 3.670\,\mathrm{m}.$$

2.7. Problem 1: floatation and stability of a sand barge

The screenshot states:

- $L = 80.000\,\text{m}$,
- $B = 20.000\,\text{m}$,
- water density $\rho_w = 1025.000\,\text{kg/m}^3$,
- sand density $\rho_s = 1600.000\,\text{kg/m}^3$,
- sand depth in the barge $= 1.750\,\text{m}$,
- neglect barge self-weight.

(1) Draught without heel or trim

The sand volume is

$$V_s = LBh = 80 \cdot 20 \cdot 1.75 = 2800.000\,\text{m}^3.$$

So its mass is

$$m = \rho_s V_s = 1600 \cdot 2800 = 4.480 \times 10^6\,\text{kg}.$$

Displacement requires

$$\rho_w \nabla = m \quad \Rightarrow \quad \nabla = \frac{m}{\rho_w} = \frac{4.48 \times 10^6}{1025} = 4370.730\,\text{m}^3.$$

For a rectangular barge,

$$\nabla = LBT,$$

so

$$T = \frac{\nabla}{LB} = \frac{4370.73}{80 \cdot 20} = 2.732\,\text{m}.$$

(2) Distance between centre of gravity and transverse metacentre

Since the load is homogeneous sand from the bottom to height $1.750\,\text{m}$,

$$KG = \frac{1.75}{2} = 0.875\,\text{m}.$$

Also,

$$KB = \frac{T}{2} = 1.366\,\text{m}.$$

The waterplane moment of area is

$$I_x = \frac{LB^3}{12} = \frac{80 \cdot 20^3}{12} = 53\,333.330\,\text{m}^4.$$

Hence,

$$BM_T = \frac{I_x}{\nabla} = \frac{53333.33}{4370.73} = 12.202\,\text{m}.$$

So

$$KM_T = KB + BM_T = 1.366 + 12.202 = 13.568\,\text{m},$$

and therefore

$$GM_T = KM_T - KG = 13.568 - 0.875 = 12.693\,\text{m}.$$

Thus the distance between the centre of gravity and the transverse metacentre is

$$\boxed{GM_T \approx 12.690\,\text{m}.}$$

(3) Restoring moment at $\varphi = 7°$

The total weight is

$$W = mg = 4.48 \times 10^6 \cdot 9.81 = 4.395 \times 10^7 \,\text{N}.$$

Using the small-angle formula,

$$GZ \approx GM_T \sin\varphi = 12.693 \sin 7° = 1.547 \,\text{m}.$$

Therefore,

$$M_R \approx WGZ = 4.39488 \times 10^7 \cdot 1.547 = 6.800 \times 10^7 \,\text{N m}.$$

So the restoring moment is approximately

$$\boxed{M_R \approx 68.000 \,\text{MN m}.}$$

(4) More precise estimate for a wall-sided barge

Using the wall-sided correction,

$$GZ = \sin\varphi \left(GM_T + \frac{1}{2} BM_T \tan^2\varphi \right).$$

With $\varphi = 7°$,

$$GZ \approx 1.558 \,\text{m}, \qquad M_R \approx WGZ \approx 68.500 \,\text{MN m}.$$

So the refined answer is only slightly larger, which makes sense because 7.000 deg is still a relatively small heel angle.

3. Resistance, Froude number, Reynolds number, and scaling

3.1. Pressure drag, viscous drag, and free-surface effects

For a fully submerged body:

- pressure drag comes from normal pressure stresses,
- viscous drag comes from tangential shear stresses.

But ships are generally *not* fully submerged. The consequence is that we also get **free-surface effects**, especially **wave-making resistance**. That is why the Froude number becomes crucial for ships.

3.2. Froude and Reynolds numbers

The two key dimensionless numbers are

$$Fr = \frac{U}{\sqrt{gL}}, \qquad Re = \frac{UL}{\nu}.$$

Physical meaning:

- Fr compares inertia to gravity effects,
- Re compares inertia to viscous effects.

3.3. Why low-speed ships do not generate much wave-making

If Fr is small, then inertia forces are small relative to gravity forces. The free surface is not strongly disturbed, so wave-making is limited.

So at low speed:

- Fr is small,
- gravity dominates over inertia at the free surface,
- the ship does not create large waves,
- viscous resistance becomes relatively more important than wave resistance.

3.4. Worked example: Fr and Re for a ship

Given:

- $L_{pp} = 250.000\,\text{m}$,
- $U = 20.000\,\text{knots}$,
- seawater at $15.000\,^\circ\text{C}$, $\nu = 1.19 \times 10^{-6}\,\text{m}^2/\text{s}$.

Convert speed:

$$U = 20 \cdot 0.514444 = 10.289\,\text{m/s}.$$

Then

$$Fr = \frac{10.289}{\sqrt{9.81 \cdot 250}} = 0.208.$$

Also,

$$Re = \frac{10.289 \cdot 250}{1.19 \times 10^{-6}} = 2.16 \times 10^{9}.$$

If transition to turbulence is assumed at $Re_x \approx 5 \times 10^5$, the nominal laminar length is

$$x = \frac{Re_x \nu}{U} = \frac{5 \times 10^5 \cdot 1.19 \times 10^{-6}}{10.289} \approx 0.058\,\text{m}.$$

So the laminar part is only a few centimetres long. That is the physical message: **real ships are almost fully turbulent over nearly all the hull.**

3.5. Geometric similarity and Froude scaling

If

$$\alpha = \frac{L_s}{L_m},$$

then geometric similarity gives

$$\alpha^2 = \frac{S_s}{S_m}, \qquad \alpha^3 = \frac{V_s}{V_m} = \frac{\nabla_s}{\nabla_m}.$$

For Froude similarity,

$$Fr_s = Fr_m \quad \Rightarrow \quad \frac{U_s}{\sqrt{gL_s}} = \frac{U_m}{\sqrt{gL_m}}.$$

Hence speed scales as

$$U \sim \sqrt{L}, \qquad \frac{U_s}{U_m} = \sqrt{\alpha}.$$

Time scales as

$$t \sim \frac{L}{U} \sim \sqrt{L}, \qquad \boxed{\frac{t_s}{t_m} = \sqrt{\alpha}.}$$

3.6. Worked example: effective power scaling

The example screenshot states:

- small ship: $L_1 = 65.000\,\mathrm{m}$, $B_1 = 6.000\,\mathrm{m}$, $T_1 = 2.000\,\mathrm{m}$,
- wetted surface $S_1 = 369.000\,\mathrm{m}^2$,
- displacement mass $= 438.000\,\mathrm{t}$,
- resistance $R_{T1} = 10.000\,\mathrm{kN}$ at $U_1 = 7.800\,\mathrm{knots}$,
- larger similar ship length $L_2 = 230.000\,\mathrm{m}$,
- form factor $k = 0.1$,
- neglect air and steering resistance.

Scale factor:

$$\alpha = \frac{230}{65} = 3.538.$$

Thus the similar dimensions are

$$B_2 = 6.0\alpha = 21.230\,\mathrm{m}, \qquad T_2 = 2.0\alpha = 7.080\,\mathrm{m}.$$

The corresponding Froude-similar speed is

$$U_2 = U_1\sqrt{\alpha} = 7.8\sqrt{3.538} = 14.670\,\mathrm{knots}.$$

To estimate effective power more realistically, split total resistance as

$$C_T = (1+k)C_F + C_R.$$

Using the ITTC-57 friction line,

$$C_F = \frac{0.075}{(\log_{10} Re - 2)^2},$$

with seawater at $15.000\,^\circ\mathrm{C}$, one obtains approximately

$$R_{T2} \approx 383.000\,\mathrm{kN}.$$

Therefore effective power is

$$P_E = R_{T2}U_2 \approx 2.890\,\mathrm{MW}.$$

If your teacher solves this example with a different extrapolation convention, the exact number may change slightly. The important exam logic is the **sequence**: geometric scaling → Froude-similar speed → friction correction with Reynolds number → effective power.

3.7. Problem 2: resistance

(1) What happens if draught increases?

In general total hull resistance increases because:

- wetted surface increases, so frictional resistance increases,
- displacement increases, which usually modifies the wave system and often increases wave-making resistance,
- the hull sinks deeper, changing pressure distribution and flow around the stern.

So the short exam answer is: **an increase in draught usually increases total resistance.**

(2) Reynolds number for $L_{pp} = 152.500\,\text{m}$ and $U = 25.000\,\text{knots}$

With

$$U = 25 \cdot 0.514444 = 12.861\,\text{m/s},$$

we get

$$Re = \frac{UL}{\nu} = \frac{12.861 \cdot 152.5}{1.19 \times 10^{-6}} = 1.65 \times 10^9.$$

(3) Which ship has the larger C_F?

At the same speed, the shorter ship has the smaller Reynolds number. Since the ITTC friction coefficient decreases with increasing Reynolds number, the shorter ship has the larger C_F.

So between $L_{pp} = 142.800\,\text{m}$ and $L_{pp} = 225.700\,\text{m}$ at the same speed, the $142.800\,\text{m}$ **ship has the larger skin-friction coefficient**.

(4) Towing-tank scale factor 29.57

Given a full-scale ship length $207.400\,\text{m}$ and a scale factor 29.57:

$$L_m = \frac{207.4}{29.57} = 7.010\,\text{m}.$$

If the ship maximum speed is $20.000\,\text{knots}$, then the model speed for Froude similarity is

$$U_m = \frac{20}{\sqrt{29.57}} = 3.680\,\text{knots} \approx 1.890\,\text{m/s}.$$

(5) Corresponding speed for a larger similar ship

If displacement mass changes from $5097.000\,\text{t}$ to $6627.000\,\text{t}$, then

$$\alpha = \left(\frac{6627}{5097}\right)^{1/3} = 1.091.$$

Hence the corresponding speed is

$$U_2 = U_1\sqrt{\alpha} = 12\sqrt{1.091} = 12.540\,\text{knots}.$$

(6) Similar ship of length $233.000\,\mathrm{m}$

Given the small ship:

- $L_1 = 64.600\,\mathrm{m}$,
- $B_1 = 6.020\,\mathrm{m}$,
- $T_1 = 1.980\,\mathrm{m}$,
- $S_1 = 369.000\,\mathrm{m}^2$,
- displacement weight $= 4.260\,\mathrm{MN}$,
- resistance $R_{T1} = 35.000\,\mathrm{kN}$ at $15.800\,\mathrm{knots}$,
- neglect form and air resistance.

Scale factor:

$$\alpha = \frac{233}{64.6} = 3.607.$$

Thus

$$B_2 = 6.02\alpha = 21.710\,\mathrm{m}, \qquad T_2 = 1.98\alpha = 7.140\,\mathrm{m}.$$

The corresponding speed is

$$U_2 = 15.8\sqrt{3.607} = 30.000\,\mathrm{knots}.$$

Using

$$C_T = C_F + C_R$$

with C_R assumed invariant under Froude similarity and C_F corrected by the ITTC line, one gets approximately

$$R_{T2} \approx 1.430\,\mathrm{MN}, \qquad P_E = R_{T2}U_2 \approx 22.100\,\mathrm{MW}.$$

4. Wave statistics and irregular seas

4.1. Why waves with periods about 5–20 s matter for floating structures

This is mainly because that period band contains much of the energy in real ocean sea states, and because it overlaps with the dynamic time scales of many floating structures.

Shorter waves often matter less because:

- they contain less total energy in many sea states,
- their loads vary rapidly along the hull and partly cancel out,
- the structure often cannot respond strongly to such high frequencies.

Very long waves matter less often because they are rarer and, for many structures, act more like slow quasi-static variations.

4.2. When does the ship respond the most?

The ship responds most when there is both:

- significant wave energy at a certain frequency, and
- a large response amplitude operator (RAO) or a natural frequency close to that frequency.

In short: **maximum response occurs when the wave spectrum and the response function overlap strongly, especially near resonance.**

4.3. Significant wave period from a sorted list

The screenshot shows 15 individual wave heights and their corresponding periods. If we define a crude significant period in the same spirit as significant wave height, we take the periods associated with the largest one-third of the waves.

The five largest wave heights correspond to periods

$$12.5,\ 13.0,\ 12.0,\ 11.2,\ 15.2\ \text{s}.$$

Therefore

$$T_{1/3} = \frac{12.5 + 13.0 + 12.0 + 11.2 + 15.2}{5} = 12.780\,\text{s}.$$

4.4. Probability of exceedance: worked example

Given:

- variance of surface elevation $m_0 = \sigma_\zeta^2 = 1.560\,\text{m}^2$,
- significant wave height $H_s = 5.000\,\text{m}$.

(1) Crest exceedance above $h_1 = 1.500\,\text{m}$

For narrow-band Gaussian seas, crest amplitudes are Rayleigh-distributed:

$$P(C > h) = \exp\left(-\frac{h^2}{2m_0}\right).$$

Hence

$$P(C > 1.5) = \exp\left(-\frac{1.5^2}{2 \cdot 1.56}\right) = 0.486.$$

So the probability is about

$$\boxed{48.6\%.}$$

(2) Wave height exceedance above H_s

For individual wave heights,

$$P(H > h) = \exp\left(-\frac{h^2}{8m_0}\right).$$

So for $h = H_s = 5.000\,\mathrm{m}$:

$$P(H > H_s) = \exp\left(-\frac{5.0^2}{8 \cdot 1.56}\right) \approx 0.135.$$

Thus

$$\boxed{P(H > H_s) \approx 13.5\%.}$$

This is a classic exam point:

- crest amplitude uses denominator $2m_0$,
- wave height uses denominator $8m_0$,
- do not mix amplitude and height formulas.

4.5. Visual/crude estimation of H_s and expected exceedances

One screenshot asks for a rough estimate from a spectrum plot only. In that kind of question, the method is more important than the exact number.

Use:

$$m_0 = \int S_\zeta(\omega)\, d\omega, \qquad H_s \approx 4\sqrt{m_0}.$$

Then estimate a representative zero-crossing or peak period from the dominant spectral frequency, compute the approximate number of waves in one hour, and multiply by the crest exceedance probability

$$P(C > h_1) = \exp\left(-\frac{h_1^2}{2m_0}\right).$$

Because the screenshot is only a visual graph with no exact data points, I would use this question mainly to remember the **procedure**, not to memorize one exact number.

5. Single-degree-of-freedom oscillations and natural frequency

5.1. Harmonic motion

If

$$x(t) = x_0 \sin(\omega t + \varepsilon),$$

then

$$\dot{x}(t) = \omega x_0 \cos(\omega t + \varepsilon), \qquad \ddot{x}(t) = -\omega^2 x_0 \sin(\omega t + \varepsilon).$$

Therefore:

- displacement amplitude $= x_0$,
- velocity amplitude $= \omega x_0$,
- acceleration amplitude $= \omega^2 x_0$.

5.2. Equation of motion with added mass

For a spring-mass-damper system with effective mass $m+a$, damping coefficient b, stiffness coefficient c, and forcing

$$F(t) = F_0 \sin(\omega t),$$

the equation of motion is

$$\boxed{(m+a)\ddot{x} + b\dot{x} + cx = F_0 \sin(\omega t).}$$

If damping is neglected ($b = 0$), the undamped natural frequency is

$$\boxed{\omega_n = \sqrt{\frac{c}{m+a}}.}$$

5.3. Natural frequency vs exciting frequency

The natural frequency ω_n is a property of the system. The exciting frequency ω is imposed by the external forcing.

They are not the same in general. But when $\omega \approx \omega_n$, resonance effects may appear and the response becomes large.

5.4. Is it reasonable that added mass and damping depend on frequency?

Yes. This is physically very reasonable.

Why?

- Added mass measures how much surrounding fluid must be accelerated together with the body.
- Radiation damping measures how much energy the oscillating body sends away into waves.
- Both processes depend on how fast the body oscillates.

So, in floating-body hydrodynamics, $a(\omega)$ and $b(\omega)$ are generally frequency-dependent.

Do not say "added mass always increases" or "always decreases" with frequency. That is too strong. The safe and correct answer is:

- they **depend on frequency**,
- damping often rises from low values, becomes important around frequencies where wave radiation is efficient, and may decrease again,
- added mass usually tends toward different asymptotic limits at low and high frequency.

The exact trend depends on body shape and motion mode.

5.5. Oscillating cylinder revisited

The screenshot later rewrites the motion equation as

$$(m+a)\ddot{z} + b\dot{z} + cz = F_a \cos(\omega t + \varepsilon_{F\zeta}).$$

The undamped natural frequency is again

$$\omega_n = \sqrt{\frac{c}{m+a}}.$$

This is exactly the same logic as above: when added mass increases, the effective inertia increases, so the natural frequency decreases.

5.6. Slide asking to "check the equations"

One screenshot shows a more advanced amplitude and phase formula and explicitly says "check the equations." Since the visible slide is cropped and depends on previous slide definitions, I would **not** memorize that formula in isolation.

For exam purposes, the safe study focus is:

- know the governing equation,
- know what the natural frequency is,
- know what resonance means,
- know that amplitude and phase both depend on frequency and damping.

6. Spectral analysis, transfer functions, and response spectra

6.1. Core formulas

If the wave spectrum is $S_\zeta(\omega)$ and the amplitude of the transfer function is $Y(\omega) = |\Phi_z(\omega)|$, then the response spectrum is

$$\boxed{S_z(\omega) = Y^2(\omega) S_\zeta(\omega).}$$

The zero-th spectral moment is

$$m_{0z} = \int_0^\infty S_z(\omega)\, d\omega.$$

For wave elevation,

$$H_{1/3} \approx 4\sqrt{m_0}.$$

For a motion amplitude like heave amplitude,

$$z_{1/3} \approx 2\sqrt{m_{0z}}.$$

The factor is 2 rather than 4 because heave is treated as an amplitude process, not crest-to-trough wave height.

6.2. Worked example: semi-submersible heave

The screenshot gives a table of ω, $S_\zeta(\omega)$, and $Y(\omega)$. With step size $\Delta\omega = 0.100\,\mathrm{rad/s}$, the wave-spectrum area is approximated by

$$\sum S_\zeta(\omega) = 12.33, \qquad m_0 \approx 0.1 \times 12.33 = 1.233\,\mathrm{m}^2.$$

So

$$H_{1/3} = 4\sqrt{1.233} = 4.440\,\mathrm{m}.$$

Next compute $Y^2(\omega)$ and multiply pointwise:

$$S_z(\omega) = Y^2(\omega) S_\zeta(\omega).$$

The screenshot values give

$$\sum S_z(\omega) \approx 2.317, \qquad m_{0z} \approx 0.1 \times 2.317 = 0.232\,\mathrm{m}^2.$$

Thus the significant heave amplitude is

$$z_{1/3} = 2\sqrt{0.232} = 0.960\,\mathrm{m}.$$

So the main answers are

$$\boxed{H_{1/3} \approx 4.440\,\mathrm{m}, \qquad z_{1/3} \approx 0.960\,\mathrm{m}.}$$

6.3. Qualitative seakeeping assessment from the overlap plot

One screenshot explicitly notes that there is no significant overlap between the wave spectrum and the transfer function.

That means:

- the sea puts most of its energy in a frequency band where the structure is not very responsive,
- therefore the response spectrum remains relatively small,
- therefore heave response is limited.

So qualitatively the semi-submersible has **good heave seakeeping characteristics in that particular sea state**. This is exactly what the overlap plot is trying to show.

6.4. Wave spectrum transformation to encounter spectrum

If a ship moves with speed U and meets waves under angle μ, then the encounter frequency is

$$\omega_e = \omega - kU\cos\mu.$$

In deep water,

$$k = \frac{\omega^2}{g},$$

so

$$\omega_e = \omega - \frac{\omega^2}{g} U\cos\mu.$$

The transformed spectrum must preserve variance, so

$$S_\zeta^*(\omega_e) = \frac{S_\zeta(\omega)}{\left|\frac{d\omega_e}{d\omega}\right|}, \qquad \frac{d\omega_e}{d\omega} = 1 - \frac{2U\omega\cos\mu}{g}.$$

6.5. Worked encounter-spectrum example

From the screenshot:

- $U = 20.000\,\text{knots} = 10.289\,\text{m/s}$,
- angle between ship and wave velocity vectors $\mu = 120°$,
- spectrum values at $\omega = 0.3, 0.4, 0.5, 0.6\,\text{rad/s}$ are $0.50, 2.00, 0.75, 0.25\,\text{m}^2\text{s/rad}$.

Since $\cos 120° = -0.5$, the encounter frequency becomes

$$\omega_e = \omega + \frac{U\omega^2}{2g}.$$

This gives approximately:

ω [rad/s]	$S_\zeta(\omega)$ [m^{2}s/rad]	ω_e [rad/s]	$S_\zeta^*(\omega_e)$ [m^{2}s/rad]
0.3	0.50	0.347	0.380
0.4	2.00	0.484	1.409
0.5	0.75	0.631	0.492
0.6	0.25	0.789	0.153

The two things to remember are:

- first transform frequency, then
- correct the spectral density by the Jacobian factor $|d\omega_e/d\omega|$.

6.6. Crude hand calculations from a spectrum and RAO

Another screenshot asks for a rough hand estimate of:

- the significant wave height,
- the probability that the heave motion exceeds a level equal to three times the standard deviation.

The slide itself estimates the wave-spectrum area as about

$$m_0 \approx 3.300\,\text{m}^2 \quad \Rightarrow \quad H_s \approx 4\sqrt{3.3} = 7.300\,\text{m}.$$

If the heave amplitude is Rayleigh distributed, the exceedance probability above $z = 3\sigma_z$ is

$$P(Z > 3\sigma_z) = \exp\left(-\frac{(3\sigma_z)^2}{2\sigma_z^2}\right) = e^{-9/2} \approx 0.011.$$

So the probability is about

$$\boxed{1.1\%.}$$

7. Weight, buoyancy, and longitudinal loads in ship strength problems

7.1. Uniform buoyant load on a rectangular barge

The screenshot gives:

- barge length $L = 100.000\,\text{m}$,
- uniformly distributed buoyant load $p = 725.000\,\text{kN/m}$,
- breadth later specified as $B = 18.000\,\text{m}$.

(a) Resultant buoyancy force

The total buoyancy force is the area under the load distribution:

$$F_B = pL = 725 \cdot 100 = 72\,500.000\,\text{kN}.$$

(b) Where does it act?

Because the load is uniform, the resultant acts at the centroid of the rectangle, i.e. at mid-length:

$$\boxed{x = L/2 = 50.000\,\text{m}.}$$

(c) Weight in static equilibrium

If the barge is in static equilibrium,

$$W = F_B = 72\,500.000\,\text{kN}.$$

(d) Draught for $B = 18.000\,\text{m}$

For a rectangular barge, buoyancy per unit length is

$$p = \rho g B T.$$

Assuming saltwater,

$$T = \frac{p}{\rho g B} = \frac{725000}{1025 \cdot 9.81 \cdot 18} \approx 4.010\,\text{m}.$$

So the draught is about $4.000\,\text{m}$.

7.2. Distribution of loads on a loaded barge

The screenshot shows four equal length segments $L/4$ with weight intensities

$$2p, \quad 5p, \quad 4p, \quad 2p,$$

where $p = 80.000\,\text{kN/m}$ and $T = 3.000\,\text{m}$.

(a) Breadth of the barge

The total distributed weight is

$$W = \frac{L}{4}(2p + 5p + 4p + 2p) = \frac{13}{4}pL.$$

If buoyancy is uniform with intensity q_B, then static equilibrium requires

$$q_B L = \frac{13}{4}pL \quad \Rightarrow \quad q_B = \frac{13}{4}p = 3.25p = 260.000\,\text{kN/m}.$$

For a rectangular barge,

$$q_B = \rho g B T.$$

Hence,

$$B = \frac{260000}{1025 \cdot 9.81 \cdot 3} \approx 8.620\,\text{m}.$$

So the breadth is approximately

$$\boxed{B \approx 8.600\,\text{m}.}$$

(b) Sketch of buoyancy load

The buoyancy load must be a **uniform upward distribution** of intensity

$$q_B = 260.000\,\text{kN/m}$$

along the whole barge length.

(c) Why is buoyancy uniform?

Because the barge is box-shaped and floats at uniform draught:

- the underwater cross-section is the same at every longitudinal position,
- therefore the displaced volume per unit length is constant,
- hence buoyancy per unit length is constant.

(d) Hogging or sagging?

The weight is more concentrated in the middle than the buoyancy is. That means the middle tends to go down relative to the ends. So the barge is in **sagging** condition.

8. What cannot be reconstructed exactly from the screenshots

8.1. MATLAB semi-sub on barge example

One screenshot from the ship-strength lecture says that a Word file on Learn contains the full example. The screenshot only gives the introduction and not the full cross-section data or final task list.

So here the correct study approach is:

- compute weight distribution of the cargo and barge,

- compute buoyancy distribution,
- form the net longitudinal load,
- integrate to get shear force and bending moment,
- compute stresses from section properties.

I cannot honestly reconstruct the missing numerical answer from the screenshot alone.

8.2. Graph-only crude estimation questions

A few screenshots are deliberately "visual" or "crude hand calculation" type questions. In those, the main exam skill is:

1. identify the correct governing formula,
2. estimate areas or peaks from the graph,
3. give an order-of-magnitude answer,
4. state clearly that it is an estimate.

Those are not questions where false precision is useful.

9. Final exam summary

For this second part, the most important reusable formulas are:

$$C_B = \frac{\nabla}{LBT}, \qquad C_M = \frac{A_M}{BT}, \qquad C_P = \frac{\nabla}{A_M L} = \frac{C_B}{C_M},$$

$$BM_T = \frac{I_x}{\nabla}, \qquad KM_T = KB + BM_T, \qquad GM_T = KM_T - KG,$$

$$GZ \approx GM \sin\varphi, \qquad M_R = WGZ,$$

$$Fr = \frac{U}{\sqrt{gL}}, \qquad Re = \frac{UL}{\nu}, \qquad C_F = \frac{0.075}{(\log_{10} Re - 2)^2},$$

$$H_s \approx 4\sqrt{m_0}, \qquad P(C > h) = \exp\left(-\frac{h^2}{2m_0}\right), \qquad P(H > h) = \exp\left(-\frac{h^2}{8m_0}\right),$$

$$(m+a)\ddot{x} + b\dot{x} + cx = F(t), \qquad \omega_n = \sqrt{\frac{c}{m+a}},$$

$$S_z(\omega) = Y^2(\omega) S_\zeta(\omega), \qquad m_{0z} = \int S_z(\omega)\, d\omega, \qquad z_{1/3} \approx 2\sqrt{m_{0z}}.$$

If you are unsure in the exam, always go back to the physical question:

- Is this a balance of forces?
- Is this a restoring moment problem?
- Is this a scaling problem under Froude similarity?
- Is this a wave-statistics problem based on spectral moments?
- Is this a response problem based on overlap between spectrum and RAO?

That physical classification usually tells you which formula to use.

41107 Marine and Ocean Engineering — Worked Examples: Probability of Exceedance and Crude Estimation

1. Cómo usar estos apuntes

Estos apuntes recogen y desarrollan los dos folios manuscritos que me has pasado, manteniendo el mismo espíritu que los apuntes de las lectures: explicación paso a paso, fórmulas con variables definidas, intuición física e ideas de examen.

Aquí hay un matiz importante: al ser apuntes a mano, hay partes donde el contexto completo no aparece en la hoja. Por eso voy a hacer dos cosas a la vez:

1. **transcribir fielmente** lo que sí se lee en la hoja;
2. **explicar con rigor** qué significa cada paso y señalar explícitamente cuando falta un dato o cuando alguna cifra no cuadra exactamente con el resto.

Eso es justo lo correcto para examen: entender el método sin inventar datos que no están escritos.

2. Qué tema están trabajando estos ejemplos

Los ejemplos corresponden al bloque de **estadística de olas irregulares**, en particular a estas ideas:

- el **momento espectral de orden cero** m_0;
- la relación entre m_0, la varianza de la elevación superficial y la altura significativa H_s;
- la **probabilidad de excedencia** de crestas o alturas de ola;
- la estimación del **número esperado de excedencias** en un cierto intervalo de tiempo.

La lógica física es esta:

> Si conoces la energía del estado de mar (por ejemplo mediante m_0 o H_s), puedes estimar qué probabilidad hay de que aparezcan olas o crestas grandes, y también cuántas de ellas esperas observar en una hora o en varias horas.

3. Formulario base que necesitas para estos ejemplos

3.1. Relación entre m_0, varianza y H_s

En teoría espectral lineal de olas, el momento espectral de orden cero es

$$m_0 = \int_0^{\infty} S(\omega)\, d\omega$$

o, si el espectro está escrito en frecuencia ordinaria,

$$m_0 = \int_0^\infty S(f)\,df.$$

Qué representa: el área bajo el espectro de oleaje. Esa área mide la energía total asociada a la elevación superficial.

Variables:

- $S(\omega)$ o $S(f)$: densidad espectral de energía;
- m_0: momento espectral de orden cero.

Relación clave:

$$\sigma^2 = m_0,$$

donde σ^2 es la varianza de la elevación de la superficie libre.

Consecuencia inmediata:

$$\sigma = \sqrt{m_0}.$$

Altura significativa:

$$H_s \approx 4\sigma = 4\sqrt{m_0}.$$

Interpretación física: cuanto mayor es el área bajo el espectro, mayor es la energía del mar, mayor es σ y por tanto mayor es H_s.

Idea de examen: la relación

$$H_s = 4\sqrt{m_0}$$

es de las que debes tener muy automatizadas.

3.2. Probabilidad de excedencia de una cresta

Si la elevación superficial es gaussiana y las crestas siguen la aproximación de Rayleigh, la probabilidad de que una cresta Z exceda un cierto nivel h_1 es

$$P(Z > h_1) = \exp\left(-\frac{h_1^2}{2\sigma^2}\right).$$

Como $\sigma^2 = m_0$, también puede escribirse como

$$P(Z > h_1) = \exp\left(-\frac{h_1^2}{2m_0}\right).$$

Qué representa: la probabilidad de que una cresta cualquiera supere el nivel vertical h_1 medido desde el nivel medio del mar.

Variables:

- Z: altura de cresta respecto al nivel medio del mar;
- h_1: nivel umbral que quieres superar;
- σ: desviación estándar de la elevación superficial;
- m_0: momento espectral de orden cero.

Intuición física: si aumentas el umbral h_1, la exponencial cae muy rápido. Por eso las olas realmente extremas son raras.

3.3. Relación entre altura de ola y cresta

En estos ejemplos se usa la relación aproximada

$$H = 2Z.$$

Qué representa: si una ola es aproximadamente simétrica respecto al nivel medio, la altura total ola-valle es el doble de la altura de cresta sobre el nivel medio.

Idea de examen: cuando te pidan pasar de una condición sobre H a una condición sobre cresta, casi siempre harás

$$H > H_* \quad \Longleftrightarrow \quad Z > \frac{H_*}{2}.$$

3.4. Número esperado de excedencias en un intervalo de tiempo

Si en un intervalo temporal observas aproximadamente N olas independientes y la probabilidad de excedencia por ola es p, el número esperado de excedencias es

$$N_{\text{exc}} = Np.$$

Y si estimas el número de olas a partir del periodo medio de cruce por cero,

$$N \approx \frac{T_{\text{tot}}}{T_z}.$$

Variables:

- T_{tot}: duración total del registro;
- T_z: periodo medio de cruce por cero;
- N: número aproximado de olas o picos observados;
- p: probabilidad de excedencia por ola.

Interpretación física: si cada ola tiene una probabilidad pequeña de superar un umbral, entonces en un registro largo puedes seguir esperando bastantes superaciones solo porque observas muchas olas.

4. Página 1 del manuscrito — Probability of exceedance

4.1. Transcripción del contenido

El título de la hoja dice:

> **Example: Probability of exceedance**

Después aparecen tres apartados numerados.

4.1.1. Ejemplo 1 — Transcripción

La hoja muestra un esquema de la superficie libre con una cresta marcada Z y un umbral horizontal h_1. Debajo aparece:

$$\begin{aligned} P(Z > h_1) &= 1 - P(Z \leq h_1) \\ &= 1 - \left(1 - \exp\left(-\frac{1}{2}\frac{h_1^2}{\sigma^2}\right)\right) \\ &= \exp\left(-\frac{1}{2}\frac{h_1^2}{\sigma^2}\right) \\ &= 63\,\%. \end{aligned}$$

4.1.2. Qué significa el ejemplo 1

Este primer ejemplo no está calculando una altura de ola completa, sino la probabilidad de que **una cresta** supere un cierto nivel h_1.

La lógica matemática es sencilla:

$$\begin{aligned} P(Z > h_1) &= 1 - P(Z \leq h_1) \\ &= 1 - F_Z(h_1), \end{aligned}$$

donde $F_Z(h_1)$ es la función de distribución acumulada de la cresta.

Si

$$F_Z(h_1) = 1 - \exp\left(-\frac{h_1^2}{2\sigma^2}\right),$$

entonces automáticamente

$$P(Z > h_1) = \exp\left(-\frac{h_1^2}{2\sigma^2}\right).$$

Intuición física: este resultado dice que las crestas pequeñas son frecuentes y las crestas muy grandes son poco frecuentes.

Comentario importante: la hoja da el resultado final **63 %**, pero el valor concreto depende del cociente h_1/σ. En la imagen se ve un texto junto al dibujo que parece indicar un valor de h_1, pero ese dato no queda suficientemente claro como para reconstruir con total seguridad el cálculo numérico exacto. Por tanto, aquí lo correcto es quedarse con el **método** y aceptar que la cifra final manuscrita es 63 %.

Idea de examen: debes saber pasar de

$$P(Z > h_1) = 1 - P(Z \leq h_1)$$

a la forma exponencial final casi sin pensar.

4.1.3. Ejemplo 2 — Transcripción

La hoja escribe:

$$H > H_s, \qquad \text{NB: } H = 2Z$$

$$P(H > H_s) = P\left(Z > \frac{1}{2}H_s\right) = \exp\left(-\frac{1}{2}\frac{\left(\frac{1}{2}H_s\right)^2}{\sigma^2}\right)$$

$$= 13.5\,\%.$$

4.1.4. Explicación paso a paso del ejemplo 2

Este ejemplo ya no pregunta por una cresta que supere un nivel cualquiera, sino por la probabilidad de que la **altura de ola** exceda la altura significativa H_s.

Paso 1: pasar de altura de ola a cresta Usamos la relación

$$H = 2Z.$$

Entonces,

$$H > H_s \iff 2Z > H_s \iff Z > \frac{H_s}{2}.$$

Paso 2: usar la fórmula de excedencia de una cresta Aplicamos

$$P(Z > h_1) = \exp\left(-\frac{h_1^2}{2\sigma^2}\right)$$

con

$$h_1 = \frac{H_s}{2}.$$

Por tanto,

$$P(H > H_s) = \exp\left(-\frac{(H_s/2)^2}{2\sigma^2}\right).$$

Paso 3: sustituir que $H_s = 4\sigma$ Si

$$H_s = 4\sigma,$$

entonces

$$\frac{H_s}{2} = 2\sigma.$$

Sustituyendo:

$$P(H > H_s) = \exp\left(-\frac{(2\sigma)^2}{2\sigma^2}\right) = \exp(-2) \approx 0.135.$$

Es decir,

$$P(H > H_s) \approx 13.5\,\%.$$

Intuición física: esto es un resultado clásico. La altura significativa no significa que la mayoría de las olas la superen; al contrario, solo una fracción relativamente pequeña la supera.

Idea de examen: este 13.5 % es casi un número de referencia que conviene recordar.

4.1.5. Ejemplo 3 — Transcripción

La hoja escribe:

$$P(H > 8\,\mathrm{m}) = \exp\left(-\frac{1}{2}\frac{\left(\frac{1}{2}\cdot 8\right)^2}{\sigma^2}\right) = 0.6\,\%.$$

4.1.6. Explicación del ejemplo 3

La estructura del cálculo es correcta. Se vuelve a usar:

1. convertir altura total en cresta, es decir $Z = H/2$;
2. aplicar la probabilidad de excedencia de una cresta.

Entonces,

$$P(H > 8\,\mathrm{m}) = P(Z > 4\,\mathrm{m}) = \exp\left(-\frac{4^2}{2\sigma^2}\right).$$

Comentario honesto importante: el manuscrito da como resultado **0.6 %**. Matemáticamente la forma es correcta, pero el valor numérico exacto depende de σ (o de m_0). En esta primera hoja ese dato no aparece de forma explícita. Por tanto, la forma del procedimiento sí puede validarse, pero la cifra final solo puede aceptarse como parte del ejemplo manuscrito, no rederivarse con total seguridad a partir de esta hoja sola.

Idea de examen: aunque falte el dato numérico, el procedimiento es siempre el mismo:

$$H > H_* \Rightarrow Z > \frac{H_*}{2} \Rightarrow P(H > H_*) = \exp\left(-\frac{(H_*/2)^2}{2\sigma^2}\right).$$

5. Página 2 del manuscrito — Crude estimation

5.1. Transcripción del contenido

El título de la segunda hoja dice:

> **Example: Crude estimation**

En la esquina superior izquierda aparece una nota que se lee como:

> **NB: True param $H_s = 4.0$ m, $T_p = 10$ s, see $S(\omega)$**

Luego la hoja escribe:

$$m_0 = 0.5 \cdot 2.25 = 1.125\,\mathrm{m}^2$$
$$H_s = 4\sqrt{m_0} = 4.2\,\mathrm{m}$$

Después aparece el bloque:

> **Exceeding a level: Probability that an arbitrary peak (crest) exceed h_1**

con la fórmula

$$P(Z > h_1) = \exp\left(-\frac{1}{2}\frac{h_1^2}{m_0}\right) = 17\,\%.$$

Más abajo pone:

> **How many peaks in 3 hours?**

y debajo:

$$N = \frac{T_{\text{tot}}}{T_z}$$

seguido de la observación:

$$\text{Generally } T_p > T_z \Rightarrow T_z \approx \frac{T_p}{1.4} = 7\,\text{s}.$$

Después aparece el cálculo:

$$N = \frac{T_{1\,\text{h}}}{T_z} = \frac{3600}{7} \approx 500.$$

Y finalmente:

$$\text{Number (expected) of exceedances: } N\,p = 500 \cdot 0.17 = 87.$$

5.2. Explicación paso a paso

5.2.1. Paso 1 — Estimación grosera de m_0 a partir del espectro

La hoja usa

$$m_0 = 0.5 \cdot 2.25 = 1.125\,\text{m}^2.$$

Qué significa: se está haciendo una estimación rápida del área bajo el espectro $S(\omega)$, es decir, de m_0.

Comentario importante: en la hoja aparece la indicación "see $S(\omega)$", pero el dibujo o gráfico del espectro no está incluido en este PDF. Por tanto, no puedo reconstruir con certeza de dónde salen exactamente los números 0.5 y 2.25. Lo único riguroso que sí puede afirmarse es que se están multiplicando dos cantidades para aproximar visualmente el área bajo el espectro.

Intuición física: esto es lo que harías en examen si te piden una *crude estimation*: mirar el espectro y aproximar su área con un triángulo, rectángulo o forma simple.

Idea de examen: si ves "crude estimation of m_0", piensa inmediatamente en **área bajo** $S(\omega)$.

5.2.2. Paso 2 — Obtener H_s a partir de m_0

La hoja sigue con

$$H_s = 4\sqrt{m_0}.$$

Sustituyendo $m_0 = 1.125\,\text{m}^2$:

$$H_s = 4\sqrt{1.125} \approx 4 \cdot 1.0607 \approx 4.24\,\text{m}.$$

Redondeando,

$$H_s \approx 4.2\,\text{m}.$$

Interpretación física: la estimación rápida del espectro ya te da una estimación rápida de la severidad del mar.

Comentario útil: la hoja dice que el valor "verdadero" era $H_s = 4.0$ m, y la estimación rápida da 4.2 m. Eso está bastante cerca, así que la crude estimation funciona razonablemente bien.

5.2.3. Paso 3 — Probabilidad de que una cresta supere h_1

La hoja escribe:

$$P(Z > h_1) = \exp\left(-\frac{1}{2}\frac{h_1^2}{m_0}\right) = 17\,\%.$$

Esto es exactamente la misma fórmula que antes, solo que sustituyendo σ^2 por m_0.

Por qué se puede hacer eso Porque

$$\sigma^2 = m_0.$$

Entonces,

$$P(Z > h_1) = \exp\left(-\frac{h_1^2}{2\sigma^2}\right) = \exp\left(-\frac{h_1^2}{2m_0}\right).$$

Comentario honesto: la hoja da directamente el resultado 17 %, pero no se ve escrito el valor concreto de h_1. Así que aquí también podemos validar el **método** y aceptar la cifra manuscrita como parte del ejemplo, pero no reconstruir completamente el dato de entrada.

5.2.4. Paso 4 — Cuántos picos hay en un registro

La hoja utiliza:

$$N = \frac{T_{\text{tot}}}{T_z}.$$

Qué significa: si cada ola dura aproximadamente T_z segundos, entonces en un tiempo total T_{tot} caben aproximadamente T_{tot}/T_z olas.

Después la hoja anota una relación práctica:

$$T_p > T_z, \qquad T_z \approx \frac{T_p}{1.4}.$$

Si $T_p = 10$ s, entonces

$$T_z \approx \frac{10}{1.4} \approx 7.14\,\text{s} \approx 7\,\text{s}.$$

Luego se calcula:

$$N = \frac{3600}{7} \approx 514 \approx 500.$$

Comentario importante: aquí hay una pequeña inconsistencia en la hoja. El texto dice "How many peaks in 3 hours?", pero la cuenta usa **3600 s**, que corresponde a **1 hora**, no a 3 horas.

Así que hay dos posibilidades:

- o bien la pregunta pretendía decir **1 hour**;
- o bien la pregunta sí era de **3 hours**, y entonces el cálculo debería haber usado $3 \cdot 3600 = 10800$ s.

Si fuera 1 hora:

$$N \approx \frac{3600}{7} \approx 500.$$

Si fuera 3 horas:

$$N \approx \frac{10800}{7} \approx 1540.$$

Idea de examen: aquí lo que importa es que controles la lógica, no memorizar el 500 sin mirar la duración total del registro.

5.2.5. Paso 5 — Número esperado de excedencias

La hoja termina con:

$$Np = 500 \cdot 0.17 = 87.$$

Esto significa:

- se toma $N \approx 500$ olas observadas;
- se toma $p = 0.17$ como probabilidad de excedencia por ola;
- el número esperado de superaciones es $500 \times 0.17 = 85$ aproximadamente.

Observación: numéricamente,

$$500 \cdot 0.17 = 85,$$

no 87. Por tanto, el 87 manuscrito parece ser un pequeño redondeo impreciso o un desliz aritmético.

Si usamos el valor más preciso $N \approx 514$:

$$514 \cdot 0.17 \approx 87.4,$$

y entonces sí sale aproximadamente 87. Esto sugiere que probablemente el autor usó el valor no redondeado antes de escribir el resultado final.

Muy importante para examen: la idea correcta es

$$\text{expected number of exceedances} = (\text{number of waves}) \times (\text{probability per wave}).$$

6. Resumen limpio de los ejemplos

6.1. Ejemplo 1 — Excedencia de una cresta

$$P(Z > h_1) = 1 - P(Z \leq h_1) = \exp\left(-\frac{h_1^2}{2\sigma^2}\right).$$

Qué debes saber: si te dan un nivel h_1 y te preguntan cuántas crestas lo superan, esta es la fórmula básica.

6.2. Ejemplo 2 — Excedencia de H_s

$$H = 2Z,$$

por tanto

$$P(H > H_s) = P\left(Z > \frac{H_s}{2}\right) = \exp\left(-\frac{(H_s/2)^2}{2\sigma^2}\right) = \exp(-2) \approx 13.5\,\%.$$

Qué debes recordar: aproximadamente el 13.5 % de las olas superan H_s.

6.3. Ejemplo 3 — Excedencia de una altura fija

$$P(H > H_*) = P\left(Z > \frac{H_*}{2}\right) = \exp\left(-\frac{(H_*/2)^2}{2\sigma^2}\right).$$

Qué debes recordar: siempre conviertes altura total en cresta antes de usar la fórmula exponencial.

6.4. Ejemplo 4 — Estimar H_s a partir de un espectro

$$m_0 = \int S(\omega)\, d\omega, \qquad H_s = 4\sqrt{m_0}.$$

Qué debes recordar: si te enseñan un espectro y te piden una estimación rápida, el objetivo es aproximar el área bajo la curva.

6.5. Ejemplo 5 — Número esperado de superaciones

$$N \approx \frac{T_{\text{tot}}}{T_z}, \qquad N_{\text{exc}} = Np.$$

Qué debes recordar: primero calculas cuántas olas hay; luego multiplicas por la probabilidad de excedencia.

7. Ideas clave de examen

1. **m_0 es el área bajo el espectro.**
2. **$\sigma^2 = m_0$ y $H_s = 4\sqrt{m_0}$.**
3. **La excedencia de crestas sigue una ley exponencial tipo Rayleigh:**

$$P(Z > h_1) = \exp\left(-\frac{h_1^2}{2m_0}\right).$$

4. **Para pasar de altura de ola a cresta, usas $H = 2Z$.**
5. **El número esperado de eventos raros puede seguir siendo grande si observas muchas olas.**
6. **En ejercicios manuscritos o de estimación rápida, es más importante entender la cadena lógica que obsesionarse con el último decimal.**

8. Resumen final de examen

Si en examen te sale un problema de este tipo, piensa en este orden:

1. **¿Te dan un espectro?** Entonces estima m_0 como área bajo $S(\omega)$.
2. **¿Te piden H_s?** Usa $H_s = 4\sqrt{m_0}$.
3. **¿Te piden una probabilidad de superar un nivel?** Usa

$$P(Z > h_1) = \exp\left(-\frac{h_1^2}{2m_0}\right).$$

4. **¿Te hablan de altura de ola en vez de cresta?** Convierte con $H = 2Z$.

5. **¿Te piden cuántas veces ocurrirá en una hora o tres horas?** Calcula primero

$$N \approx \frac{T_{\mathrm{tot}}}{T_z},$$

y luego multiplica por la probabilidad:

$$N_{\mathrm{exc}} = Np.$$

La intuición final es muy importante: un evento raro por ola puede dejar de parecer tan raro cuando observas cientos o miles de olas.

Transcription of example exercises and MATLAB functionality

41107 Marine and Ocean Engineering

This document is a clean transcription of the uploaded example sheets, screenshots, handwritten solutions, and MATLAB files. I keep the original exercise titles as closely as possible. Where a handwritten result or screenshot is inconsistent with the stated data, I keep the title and the intent of the exercise, but I also flag the correction explicitly.

Useful cross-references to the PDFs already generated earlier:

- `homework_notes_first_part_41107.pdf` for the first part of the course.
- `homework_notes_second_part_41107.pdf` for the second part of the course.
- `lecture_examples_wave_statistics_notes.pdf` for the wave-statistics formulas used in the exceedance examples.
- `lecture12_strength_calculations_of_ships_notes.pdf` for the ship-strength background used in the bending-moment example.

Example: Probability of exceedance

Transcribed statement. Consider a wave system represented by sea state code 6. For the time history recorded (a total of 23.5 minutes), the variance of the surface elevation was $1.56\,\mathrm{m}^2$, and the significant wave height was $5.0\,\mathrm{m}$. Find:

1. the probability that a wave crest exceeds the level $h_1 = 1.5\,\mathrm{m}$,
2. the probability that the wave height exceeds the significant wave height,
3. the probability that the wave height exceeds $8.0\,\mathrm{m}$.

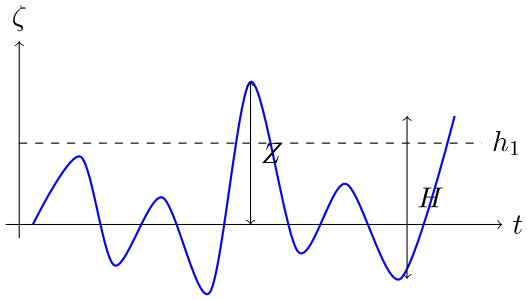

Figure 1: *
Simple sketch: crest amplitude Z and wave height H.

The key formulas are:

$$P(Z > h) = \exp\left(-\frac{h^2}{2m_0}\right), \qquad P(H > h) = \exp\left(-\frac{h^2}{8m_0}\right).$$

Here $m_0 = \sigma_\zeta^2 = 1.56\,\mathrm{m}^2$.

1) Crest exceedance above $h_1 = 1.5\,\mathrm{m}$.

$$P(Z > 1.5) = \exp\left(-\frac{1.5^2}{2 \cdot 1.56}\right) = 0.486.$$

So the correct probability is

$$\boxed{P(Z > 1.5) \approx 48.6\%.}$$

2) Wave height exceedance above $H_s = 5.0\,\mathrm{m}$.

$$P(H > H_s) = \exp\left(-\frac{5.0^2}{8 \cdot 1.56}\right) = 0.135.$$

Therefore

$$\boxed{P(H > H_s) \approx 13.5\%.}$$

3) Wave height exceedance above 8.0 m.

$$P(H > 8) = \exp\left(-\frac{8.0^2}{8 \cdot 1.56}\right) = 0.00595.$$

Hence

$$\boxed{P(H > 8) \approx 0.60\%.}$$

The handwritten page writes about **63%** for part 1. That value is not consistent with the stated variance $1.56\,\mathrm{m}^2$. Using the standard crest-exceedance formula gives **48.6%**. Parts 2 and 3 are consistent with the formulas.

Example: Visual/crude estimation....

Transcribed task.

- Determine H_s.
- How many exceedances of the level $h_1 = 2.0\,\mathrm{m}$ can be expected in a period of one hour?

From the handwritten note, the intended rough estimate is:

- true parameters noted on the page: approximately $H_s = 4.0\,\mathrm{m}$ and $T_p \approx 10\,\mathrm{s}$,
- area under the spectrum estimated visually as

$$m_0 \approx 0.5 \times 2.25 = 1.125\,\mathrm{m}^2,$$

- hence

$$H_s \approx 4\sqrt{m_0} = 4\sqrt{1.125} = 4.24\,\mathrm{m} \approx \boxed{4.2\,\mathrm{m}}.$$

For crest exceedance of level $h_1 = 2.0\,\mathrm{m}$:

$$P(Z > h_1) = \exp\left(-\frac{h_1^2}{2m_0}\right) = \exp\left(-\frac{2.0^2}{2 \cdot 1.125}\right) = 0.169.$$

So the exceedance probability is approximately

$$\boxed{P(Z > 2.0) \approx 17\%.}$$

To estimate the number of peaks in one hour, the handwritten note uses an approximate average zero-crossing period

$$T_z \approx 7\,\text{s}.$$

Therefore

$$N \approx \frac{3600}{7} \approx 514 \approx 500 \text{ peaks per hour.}$$

Expected number of exceedances:

$$N\,P \approx 500 \times 0.17 \approx \boxed{85\text{–}87}.$$

The handwritten note contains a small wording inconsistency: it mentions **3 hours**, but the actual count uses **3600 s**, which corresponds to **1 hour**. The numerical result ≈ 87 is therefore a one-hour estimate, not a three-hour estimate.

Example

Transcribed statement. A ship floating freely in saltwater is 150 m long, 22 m beam and 9 m draught and has a mass displacement of 24 000 tons. The midship wetted area is 180 m^2. Find the block, prismatic and midship area coefficients.

Take seawater density as $\rho = 1025\,\text{kg/m}^3$. The displaced volume is

$$\nabla = \frac{\Delta}{\rho} = \frac{24{,}000 \times 1000}{1025} = 23\,415\,\text{m}^3.$$

Then

$$C_B = \frac{\nabla}{LBT} = \frac{23415}{150 \cdot 22 \cdot 9} = \boxed{0.79}.$$

Using $A_M = 180\,\text{m}^2$,

$$C_P = \frac{\nabla}{A_M L} = \frac{23415}{180 \cdot 150} = \boxed{0.87}.$$

And

$$C_M = \frac{A_M}{BT} = \frac{180}{22 \cdot 9} = \boxed{0.91}.$$

Example: Distances for a rectangular barge

Transcribed statement. A rectangular box of length L, beam B, and depth D, floats at a uniform draught T. Deduce expressions for KB, BM_T and KM_T in terms of the principal dimensions. If the beam is 9 m, what must be the draught for the metacentre to lie in the waterplane?

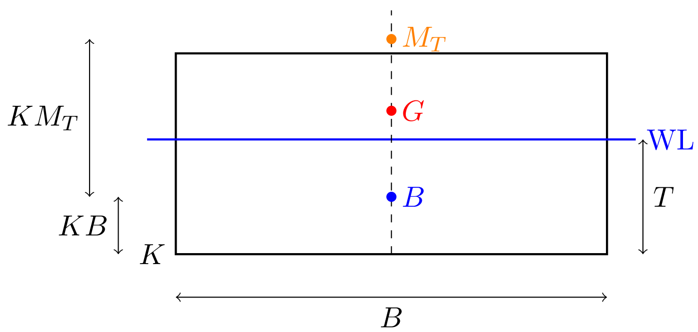

Figure 2: *
Rectangular-barge sketch for KB, BM_T and KM_T.

For a rectangular barge:

$$KB = \frac{T}{2}, \qquad \nabla = LBT, \qquad I_x = \frac{1}{12}LB^3.$$

Thus

$$BM_T = \frac{I_x}{\nabla} = \frac{B^2}{12T}.$$

Hence

$$KM_T = KB + BM_T = \frac{T}{2} + \frac{B^2}{12T}.$$

If the metacentre is to lie in the waterplane, then $KM_T = T$, so

$$\frac{T}{2} + \frac{B^2}{12T} = T.$$

This gives

$$\frac{B^2}{12T} = \frac{T}{2} \quad \Rightarrow \quad T^2 = \frac{B^2}{6} \quad \Rightarrow \quad T = \frac{B}{\sqrt{6}}.$$

For $B = 9\,\mathrm{m}$:

$$T = \frac{9}{\sqrt{6}} = \boxed{3.67\,\mathrm{m}}.$$

Example (MATLAB)

Transcribed statement. A ship has dimensions $L_{pp} = 144.0\,\mathrm{m}$ and $L_{oa} = 165.6\,\mathrm{m}$. The waterline breadths ($B = 2y$) are given in the table on the slide. Questions:

1. Calculate the waterplane area.
2. Calculate the geometrical centre of the waterplane.

This is exactly what the MATLAB file `NumIntegration.m` does. It stores the tabulated waterline half-breadths $y(x)$ and applies trapezoidal integration.

With the values from the script:

$$A_W = \int B(x)\,dx = \int 2y(x)\,dx \approx \boxed{5246.64\,\mathrm{m}^2}.$$

The geometric centre of the waterplane is obtained from the first area moment:

$$x_F = \frac{\int x\,B(x)\,dx}{A_W}.$$

Using the script data gives

$$\boxed{x_F \approx 73.90\,\mathrm{m}}$$

from the aft-most tabulated origin $x' = 0$, or equivalently

$$\boxed{x_F \approx 52.30\,\mathrm{m}}$$

forward of the aft perpendicular reference $x = 0$ used in the table.

What the code does. `NumIntegration.m` defines the arrays `xx` and `y`, computes the waterplane area with `trapz(xx,2*y)`, then computes the first area moment `trapz(xx,xx.*(2*y))` and divides by the area to obtain the geometric centre. In other words, it is a direct numerical version of the formulas you would write by hand.

Floatation - The physics behind...

A ship floats because the total hydrostatic pressure force balances the ship weight. The two resultants are:

- the weight acting downward through the centre of gravity G,
- the buoyant force acting upward through the centre of buoyancy B.

If the ship is upright and in equilibrium, these two forces are collinear. Once the ship heels, the underwater geometry changes, the centre of buoyancy shifts, and a restoring or overturning moment appears.

Stability of ships

When a ship heels, the centre of buoyancy moves toward the side with the larger immersed volume. The key question is whether the line of action of buoyancy moves in such a way that it creates a restoring couple with the ship weight.

A convenient small-angle measure is the transverse metacentric height:

$$GM_T = KB + BM_T - KG.$$

If $GM_T > 0$, the ship is initially stable. A higher centre of gravity reduces GM_T and therefore reduces stability, but it does not automatically imply capsize.

The hydrostatic restoring moment, cont'd

For a wall-sided vessel at small heel angle ϕ,

$$M_R \approx \rho g \nabla GM_T \sin\phi.$$

The metacentric radius is

$$BM_T = \frac{I_x}{\nabla},$$

where I_x is the transverse second moment of the waterplane area. This is the formula highlighted on the slide.

Problem 1: Floatation and stability

Transcribed statement. A barge with dimensions $L = 80\,\mathrm{m}$, $B = 20\,\mathrm{m}$ floats in still water ($\rho = 1025\,\mathrm{kg/m^3}$). The barge can be assumed to be homogeneously loaded with sand ($\rho = 1600\,\mathrm{kg/m^3}$). The sand is filled to a height $1.75\,\mathrm{m}$ above the bottom of the barge. Neglect the barge's own mass.

Find:

1. the draught T without heel or trim,
2. the distance between the centre of gravity and the transverse metacentre,
3. the restoring moment for a heel angle of 7° to starboard,
4. a more precise estimate of the restoring moment (wall-sided barge).

This is the exercise solved by `StabilityExercise.m`.

1) Draught. The sand mass is

$$W = LBD_{sand}\rho_s = 80 \cdot 20 \cdot 1.75 \cdot 1600 = 4.48 \times 10^6 \text{ kg}.$$

The required draught follows from buoyancy balance:

$$T = \frac{W}{\rho_w LB} = \frac{4.48 \times 10^6}{1025 \cdot 80 \cdot 20} = \boxed{2.73\,\text{m}}.$$

2) Distance between centre of gravity and transverse metacentre. Displacement volume:

$$\nabla = LBT = 80 \cdot 20 \cdot 2.73 = 4371\,\text{m}^3.$$

Centre of gravity of the sand:

$$KG = \frac{1.75}{2} = 0.875\,\text{m}.$$

Centre of buoyancy:

$$KB = \frac{T}{2} = 1.366\,\text{m}.$$

Waterplane second moment:

$$I = \frac{1}{12}LB^3 = \frac{1}{12} \cdot 80 \cdot 20^3 = 5.33 \times 10^4\,\text{m}^4.$$

Thus

$$BMT = \frac{I}{\nabla} = 12.2\,\text{m}, \qquad GMT = (KB + BMT) - KG = \boxed{12.7\,\text{m}}.$$

3) Restoring moment at 7°.

$$M_{R,0} = \rho_w g \nabla\, GMT \sin\phi = 1025 \cdot 9.81 \cdot 4371 \cdot 12.7 \cdot \sin 7^\circ.$$

This gives

$$\boxed{M_{R,0} \approx 68.0\,\text{MNm}}.$$

4) More precise wall-sided estimate. The code uses the wall-sided correction:

$$M_{R,1} = \rho_w g \nabla \sin\phi \left(GMT + \frac{1}{2} BMT \tan^2\phi \right).$$

This gives

$$\boxed{M_{R,1} \approx 68.5\,\text{MNm}}.$$

What the MATLAB code does. `StabilityExercise.m` follows the same steps as the hand solution: it first computes the draught from buoyancy balance, then calculates ∇, KG, KB, I, BMT, and GMT, and finally evaluates the restoring moment once with the small-angle metacentric formula and once with the wall-sided correction.

Example: Froude and Reynolds numbers for a ship

Transcribed statement. A ship with $L_{pp} = 250\,\text{m}$ is traveling at a speed of 20 knots. Calculate the Froude and Reynolds numbers for this ship and speed. Saltwater at temperature 15 °C is assumed. Assuming that the transition to turbulent flow happens for $Re \sim 5 \times 10^5$, determine the nominal length of laminar flow.

Take $\nu = 1.19 \times 10^{-6}\,\mathrm{m^2/s}$. Convert speed:

$$U = 20 \cdot 0.514444 = 10.289\,\mathrm{m/s}.$$

Then

$$Fr = \frac{U}{\sqrt{gL}} = \frac{10.289}{\sqrt{9.81 \cdot 250}} = \boxed{0.208 \approx 0.21}.$$

Also

$$Re = \frac{UL}{\nu} = \frac{10.289 \cdot 250}{1.19 \times 10^{-6}} = \boxed{2.16 \times 10^9}.$$

Nominal laminar length from $Re_x \approx 5 \times 10^5$:

$$x = \frac{Re_x \nu}{U} = \frac{5 \times 10^5 \cdot 1.19 \times 10^{-6}}{10.289} = \boxed{0.058\,\mathrm{m}}.$$

So the nominal laminar length is only about 6 cm: for a full-scale ship, the boundary layer is essentially turbulent over almost the whole hull.

Example: Effective power

Transcribed statement. A ship of length 65 m, breadth 6.0 m, draught 2.0 m, wetted surface 369 m^2, and mass displacement 438 tons has a resistance of 10 kN at speed 7.8 knots. Deduce the dimensions of a ship of similar form but with length 230 m. What is the effective power of the larger ship when it sails at the corresponding speed? Assume the form factor $k = 0.1$ and neglect air and steering resistance.

Geometric scale factor:

$$\alpha = \frac{230}{65} = 3.538.$$

Hence

$$B_2 = \alpha B_1 = 3.538 \cdot 6.0 = \boxed{21.2\,\mathrm{m}}, \qquad T_2 = \alpha T_1 = 3.538 \cdot 2.0 = \boxed{7.08\,\mathrm{m}}.$$

At corresponding Froude-scaled speed,

$$V_2 = \sqrt{\alpha}\, V_1 = \sqrt{3.538} \cdot 7.8 = \boxed{14.67\,\mathrm{knots}}.$$

Also

$$\Delta_2 = \alpha^3 \Delta_1 = \boxed{1.94 \times 10^4\,\mathrm{tons}}, \qquad S_2 = \alpha^2 S_1 = \boxed{4620\,\mathrm{m^2}}.$$

For ship 1,

$$Re_1 = \frac{U_1 L_1}{\nu} = 2.19 \times 10^8, \qquad C_{F1} = \frac{0.075}{(\log_{10} Re_1 - 2)^2} = 1.865 \times 10^{-3}.$$

Total resistance coefficient from the measured resistance:

$$C_{T1} = \frac{R_1}{\frac{1}{2}\rho U_1^2 S_1} = 3.284 \times 10^{-3}.$$

Thus the residual coefficient is

$$C_R = C_{T1} - C_{F1}(1 + k) = 1.232 \times 10^{-3}.$$

For ship 2,

$$Re_2 = 1.46 \times 10^9, \qquad C_{F2} = 1.461 \times 10^{-3}.$$

So

$$C_{T2} = C_{F2}(1 + k) + C_R = 2.840 \times 10^{-3}.$$

Then

$$R_2 = \tfrac{1}{2}\rho U_2^2 S_2 C_{T2} = \boxed{383\,\mathrm{kN}}.$$

Effective power:

$$P_E = R_2 U_2 = 383 \times 10^3 \cdot (14.67 \cdot 0.514444) = \boxed{2.89\,\mathrm{MW}}.$$

Problem 2: Resistance

Transcribed statement. Answer the following questions; if needed you can assume saltwater at 15 °C.

1. What would happen to the total hull resistance of a ship if the ship's draught (i.e. displacement) were to increase?
2. A ship with $L_{pp} = 152.5\,\text{m}$ is traveling at a speed of 25 knots. Calculate the Reynolds number for this ship and speed.
3. A ship with $L_{pp} = 142.8\,\text{m}$ and another ship with $L_{pp} = 225.7\,\text{m}$ are steaming together at a speed of 23 knots. Which ship will have a greater skin friction coefficient C_F?
4. A new class of supply ship is being tested in the towing tank. The ship has a length of 207.4 m and the model is built to a scale factor of 29.57. (a) What length is the model? (b) The ship has a maximum speed of 20 knots. What speed must the model be towed at in order to achieve partial dynamic similarity?
5. A ship of 5097 t mass displacement is driven at a speed of 12 knots. A ship of 6627 t of similar form is being designed. At what corresponding speed of the larger ship should its performance be compared with the 5097 t ship? [Partial dynamic similarity is assumed]
6. A ship of length 64.6 m, 6.02 m beam, 1.98 m draught, wetted surface $369\,\text{m}^2$ and displacement 4.26 MN has a resistance of 35 kN at 15.8 knots. Deduce the dimensions and effective power of a ship of similar form 233 m long at the corresponding speed. Neglect form and air resistance.

Q1. Increase in draught. If the draught increases, the wetted surface usually increases and the ship also tends to displace more water. Therefore the total hull resistance increases.

Q2. Reynolds number. Using $U = 25 \times 0.514444 = 12.861\,\text{m/s}$ and $\nu = 1.19 \times 10^{-6}\,\text{m}^2/\text{s}$:

$$Re = \frac{U L_{pp}}{\nu} = \frac{12.861 \cdot 152.5}{1.19 \times 10^{-6}} = \boxed{1.65 \times 10^9}.$$

Q3. Which ship has the greater skin-friction coefficient? At the same speed, the shorter ship has the smaller Reynolds number. Since

$$C_F = \frac{0.075}{(\log_{10} Re - 2)^2},$$

C_F decreases when Re increases. Therefore the smaller ship has the larger C_F. Numerically,

$$Re_{142.8} \approx 1.42 \times 10^9, \qquad Re_{225.7} \approx 2.24 \times 10^9,$$

so the **142.8 m ship** has the greater skin-friction coefficient.

Q4. Model length and towing speed.

$$L_m = \frac{L_s}{\alpha} = \frac{207.4}{29.57} = \boxed{7.01\,\text{m}}.$$

For Froude similarity,

$$V_m = \frac{V_s}{\sqrt{\alpha}} = \frac{20}{\sqrt{29.57}} = \boxed{3.68\,\text{knots}} = \boxed{1.89\,\text{m/s}}.$$

Q5. Corresponding speed for the larger ship. Scale factor from similar displacement:

$$\alpha = \left(\frac{6627}{5097}\right)^{1/3} = 1.092.$$

Hence

$$V_2 = \sqrt{\alpha}\, V_1 = \sqrt{1.092} \cdot 12 = \boxed{12.54\,\text{knots}}.$$

Q6. Larger similar ship: dimensions and effective power. Geometric scale factor:

$$\alpha = \frac{233}{64.6} = 3.607.$$

So

$$B_2 = 6.02 \cdot 3.607 = \boxed{21.71\,\text{m}}, \qquad T_2 = 1.98 \cdot 3.607 = \boxed{7.14\,\text{m}}.$$

The corresponding speed is

$$V_2 = V_1\sqrt{\alpha} = 15.8\sqrt{3.607} = \boxed{30.0\,\text{knots}}.$$

The larger displacement force scales as α^3:

$$\Delta_2 \approx \boxed{200\,\text{MN}}.$$

The handwritten solution gives

$$R_{tot,2} \approx \boxed{1.43\,\text{MN}}.$$

Therefore the effective power becomes

$$P_{E,2} = R_{tot,2} V_2 \approx \boxed{22.1\,\text{MW}}.$$

Example: Weight and buoyancy

Transcribed statement. Consider a 100 m long rectangular box-shaped barge. The barge is experiencing a uniformly distributed buoyant load of $p = 725\,\text{kN/m}$. Find:

a. the resultant buoyant force,

b. the point at which the resultant buoyant force is acting,

c. the barge weight if it is in static equilibrium,

d. the draught if the breadth is $B = 18\,\text{m}$.

The resultant buoyant force is simply the area under the uniform load:

$$F_B = pL = 725 \cdot 100 = \boxed{72\,500\,\text{kN}}.$$

Since the load is uniform, it acts at midship:

$$\boxed{x = L/2 = 50\,\text{m}}.$$

If the barge is in static equilibrium, its weight must equal the buoyancy:

$$\boxed{W = 72\,500\,\text{kN}}.$$

For a rectangular barge,

$$p = \rho g B T.$$

So with $B = 18\,\text{m}$,

$$T = \frac{p}{\rho g B} = \frac{725000}{1025 \cdot 9.81 \cdot 18} = \boxed{4.01\,\text{m}}.$$

Example: Distribution of loads

Transcribed statement. A loaded box-shaped (rectangular) barge in hydrostatic equilibrium has the weight (including cargo) distribution shown on the slide. The barge has draught $T = 3.0\,\mathrm{m}$. The load intensity is $p = 80\,\mathrm{kN/m}$. Find:

a. the barge breadth B,

b. sketch the uniformly distributed buoyancy load required for static equilibrium,

c. explain why the buoyancy load can be considered uniform,

d. determine whether the barge is in hogging or sagging in still water.

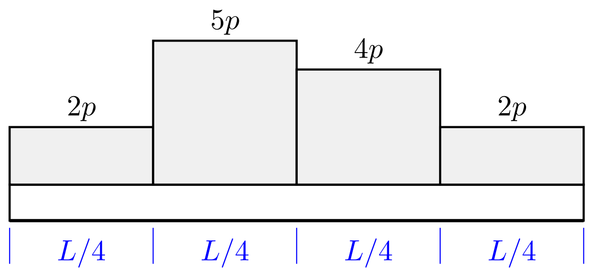

Figure 3: *
Recreated sketch of the four quarter-length load blocks.

The total weight is

$$W = \frac{L}{4}(2p + 5p + 4p + 2p) = \frac{13}{4}pL.$$

Static equilibrium requires uniform buoyancy intensity

$$q_B = \frac{W}{L} = \frac{13}{4}p = 3.25p = 260\,\mathrm{kN/m}.$$

Since for a rectangular barge

$$q_B = \rho g B T,$$

we get

$$B = \frac{260000}{1025 \cdot 9.81 \cdot 3} = \boxed{8.62\,\mathrm{m}}.$$

The buoyancy load is uniform because the barge is box-shaped and floats at uniform draught. That means the underwater cross-section is the same at every longitudinal position, so the displaced volume per unit length is constant.

Because the weight is more concentrated in the middle than the buoyancy is, the barge is in

$$\boxed{\text{sagging}}.$$

Example (made via Matlab), see Word file available from Learn

The screenshot points to a larger ship-strength example that is fully written in the separate Word file. That Word file is the document **Example: Transportation of a semi-submersible**, transcribed in the next section.

The small handwritten summary page contains the following numerical points:

1. cross-sectional steel area
$$A = 2D\,t + 3Dt + 2B\,2t + 2 \cdot 9 \cdot 2t = \boxed{4.29\,\mathrm{m}^2},$$
2. steel volume
$$V_s \approx A\,L + 2BD\,2t = \boxed{730\,\mathrm{m}^3},$$
3. total transported mass
$$\Delta \approx W_{steel} + W_{semi-sub} + W_{ballast} = \boxed{70873 \text{ tons}},$$
4. draught
$$T = \frac{\Delta}{\rho LB} = \boxed{10.8\,\mathrm{m}},$$
5. for uniform total load, the still-water vertical bending moment is
$$\boxed{VBM_{SW} = 0},$$
6. neutral axis from keel
$$z_n = \frac{S_y}{A} = \boxed{8.81\,\mathrm{m}},$$
7. second moment of area about the neutral axis
$$I_{y,n} = \boxed{231.9\,\mathrm{m}^4}.$$

For the wave-induced part, the handwritten summary itself says "See Matlab script" because the response-spectrum calculation depends on the external RAO data file. So the correct way to study this example is together with the Word statement and the partial script `BendingMomentBarge.m`.

Example: Transportation of a semi-submersible

Transcribed statement. A semi-submersible is transported on a barge-like ship. A simplified strength analysis must be made to ensure the capacity of the barge.

The mass of the semi-sub can be assumed to be $M = 30{,}000$ tons. The barge itself is made of steel with density $\rho_s = 8050\,\mathrm{kg/m^3}$, and it has main dimensions: length $L = 160\,\mathrm{m}$ and breadth $B = 40\,\mathrm{m}$. It is assumed that the barge has the same equivalent cross section over its length. The ends (forward and aft) of the barge are taken as solid plates with thickness $2t$, where $t = 15\,\mathrm{mm}$. In addition to the weight of steel, the barge's total weight is made up of equipment and ballast. The weight of equipment and ballast is 35,000 tons, and it is distributed so that the total weight per unit length from steel, equipment, ballast, and semi-sub is uniformly distributed along the length.

During transportation, the speed is $U = 8\,\mathrm{knots}$. Head sea ($\beta = 180°$) is considered only, and the limiting sea state is described by a Bretschneider wave spectrum with significant wave height $H_s = 2.5\,\mathrm{m}$ and zero-upcrossing period $T_z = 8.5\,\mathrm{s}$. The amplitude of the transfer function for the vertical bending moment amidships has been pre-computed for the condition $U = 8\,\mathrm{knots}$ and $\beta = 180°$. The output is available from the file `VBM_RAO.xls`; the absolute and encounter frequencies are given in rad/s, and the modulus is in Nm/m.

The tasks are:

1. What is the draught of the barge during transportation?
2. Calculate the still-water bending moment amidships of the barge.

3. Use the available transfer function and plot the resulting response spectrum of the wave-induced bending moment amidships, assuming the limiting sea state described above.

4. Calculate the most probable largest wave-induced bending moment during transportation, if the limiting sea state occurs during a three-hours period.

5. Determine the cross-sectional parameters (neutral axis and moment of inertia).

6. Compute the corresponding stresses in the keel and deck plates, considering the total bending moment amidships (still-water plus wave-induced moments). Is it safe to use the barge for transporting the semi-sub?

What can be stated numerically from the uploaded script material.

- Draught of barge: $\boxed{T \approx 10.80\,\mathrm{m}}$.
- Still-water bending moment: $\boxed{M_{SW} = 0}$, because both total weight and buoyancy are assumed uniformly distributed.
- Most probable largest wave-induced bending moment: the partial script comment reports

$$\boxed{MPL \approx 1.81\,\mathrm{GNm}}.$$

- Neutral axis from keel: $\boxed{z_n \approx 8.81\,\mathrm{m}}$.
- Second moment of area: $\boxed{I_{y,n} \approx 231.9\,\mathrm{m}^4}$.
- Extreme bending stresses reported by the script:

$$\sigma_{deck} \approx \boxed{71.9\,\mathrm{MPa}}, \qquad \sigma_{keel} \approx \boxed{-68.9\,\mathrm{MPa}}.$$

Since ordinary ship steel has yield strength around 235 MPa, and often even higher-strength steel is used, the script conclusion is that the barge is **safe** for this simplified check.

What the partial MATLAB script does. The file `BendingMomentBarge.m` is structured question by question:

1. it defines constants and the barge geometry,
2. computes the draught from the total transported mass,
3. sets the still-water bending moment to zero because the load and buoyancy are both uniform,
4. reads the pre-computed RAO from `VBM_RAO.xls`,
5. builds a Bretschneider wave spectrum from H_s and T_z,
6. multiplies spectrum and squared RAO to form the response spectrum,
7. integrates spectral moments to obtain the most probable largest bending moment,
8. computes neutral axis and second moment of area,
9. converts bending moment into deck and keel stresses.

Some lines are intentionally missing in the uploaded partial script, so the logic is recoverable even though the exact RAO-based plot cannot be recomputed without the Excel file.

Example: Harmonic motion and the natural frequency

Transcribed statement. Consider the motion

$$x(t) = x_0 \sin(\omega t + \varepsilon).$$

Determine the velocity and the acceleration, and specify the amplitudes. Then consider a spring-mass-damper system with effective mass $m + a$, damping coefficient b, stiffness coefficient c, and forcing $F(t) = F_0 \sin(\omega t)$. Write the equation of motion, find the natural undamped frequency for $b = 0$, and explain the difference between the natural frequency and the exciting frequency.

Differentiate the motion:

$$\dot{x}(t) = \omega x_0 \cos(\omega t + \varepsilon), \qquad \ddot{x}(t) = -\omega^2 x_0 \sin(\omega t + \varepsilon).$$

So the amplitudes are:

$$\boxed{x_a = x_0}, \qquad \boxed{\dot{x}_a = \omega x_0}, \qquad \boxed{\ddot{x}_a = \omega^2 x_0}.$$

Equation of motion:

$$\boxed{(m + a)\ddot{x} + b\dot{x} + cx = F_0 \sin(\omega t).}$$

If $b = 0$, the undamped natural frequency is obtained from the homogeneous equation:

$$(m + a)\ddot{x} + cx = 0.$$

Assuming harmonic free motion $x \propto e^{i\omega_n t}$ gives

$$\boxed{\omega_n = \sqrt{\frac{c}{m + a}}.}$$

The difference between the two frequencies is conceptual:

- ω_n is a property of the system itself,
- ω is the forcing frequency imposed from outside.

When ω is close to ω_n, resonance-type amplification may occur.

Homework: Is it reasonable to assume that added mass and damping depend on the wave (oscillating) frequency? Will they increase/decrease with frequency?

Yes, it is reasonable. In fact, for floating-body hydrodynamics it is the normal situation.

- **Added mass** depends on frequency because the amount of fluid that is effectively accelerated together with the body depends on the oscillation pattern.
- **Radiation damping** depends on frequency because the ability of the body to radiate waves depends on how fast it oscillates.

The safe exam answer is not to claim one universal monotonic trend. The exact variation depends on the body geometry and motion mode. What is important is to state that both are generally frequency-dependent and that this affects the response amplitude and phase.

A simplified example: An oscillating cylinder... revisited

The later slide rewrites the linear equation of motion as

$$(m+a)\ddot{z}+b\dot{z}+cz=F_a\cos(\omega t+\varepsilon_{F\zeta}).$$

The undamped natural frequency is again

$$\boxed{\omega_n=\sqrt{\frac{c}{m+a}}.}$$

The physical message is exactly the same: if the effective inertia increases because of added mass, then the natural frequency decreases.

Spectral analysis: An example

Transcribed statement. The table on the slide shows corresponding values between the wave frequency ω and a wave spectrum $S_\zeta(\omega)$ together with amplitudes of the heave transfer function $Y(\omega)$ of a semi-submersible. Tasks:

a) estimate the significant wave height $H_{1/3}$,

b) assess qualitatively the seakeeping characteristics,

c) determine the significant heave amplitude $z_{1/3}$.

Using the table spacing $\Delta\omega = 0.1\,\mathrm{rad/s}$, the zero-th moment of the wave spectrum is approximated as

$$m_0\approx\Delta\omega\sum S_\zeta(\omega)=0.1\times 12.33=1.233\,\mathrm{m}^2.$$

Hence

$$H_{1/3}\approx 4\sqrt{m_0}=4\sqrt{1.233}=\boxed{4.44\,\mathrm{m}}.$$

Now form the response spectrum

$$S_z(\omega)=Y^2(\omega)S_\zeta(\omega).$$

Using the tabulated values gives

$$\sum S_z(\omega)\approx 2.317,\qquad m_{0z}\approx 0.1\times 2.317=0.232\,\mathrm{m}^2.$$

For heave amplitude,

$$z_{1/3}\approx 2\sqrt{m_{0z}}=2\sqrt{0.232}=\boxed{0.96\,\mathrm{m}}.$$

Qualitative seakeeping assessment. The slide notes that there is no significant overlap between the wave spectrum and the transfer function. This means the structure is not very responsive in the frequency band where the sea has most of its energy, so the heave response remains modest. In this particular sea state, the semi-submersible therefore has good heave seakeeping characteristics.

Example, cont'd

This sequence of slides is essentially the worked continuation of the previous spectral-analysis example. The computation is:

$$S_z(\omega)=Y^2(\omega)S_\zeta(\omega).$$

The red and blue curves on the slide are the wave spectrum and the transfer function amplitude; the yellow curve is the resulting response spectrum.

The numerical take-away is:

$$\boxed{m_{0z} \approx 0.232\,\mathrm{m}^2, \qquad z_{1/3} \approx 0.96\,\mathrm{m}.}$$

Example: Wave spectrum transformation

Transcribed statement. A wave spectrum $S_\zeta(\omega)$ of an irregular long-crested wave system, as measured at a fixed point, is given by:

ω [rad/s]	0.3	0.4	0.5	0.6
$S_\zeta(\omega)$ [m^{2}s/rad]	0.50	2.00	0.75	0.25

A ship heads into this wave system at 20 knots, encountering the waves at an angle of 120°. Calculate the encounter spectrum $S^*_\zeta(\omega_e)$ that a moving wave probe would measure.

Convert ship speed:

$$U = 20 \times 0.514444 = 10.289\,\mathrm{m/s}.$$

In deep water,

$$\omega_e = \omega - \frac{\omega^2}{g} U \cos\beta, \qquad \frac{d\omega_e}{d\omega} = 1 - \frac{2U\omega}{g}\cos\beta.$$

Since $\cos 120° = -0.5$,

$$\omega_e = \omega + \frac{U\omega^2}{2g}, \qquad \frac{d\omega_e}{d\omega} = 1 + \frac{U\omega}{g}.$$

Variance preservation gives

$$S^*_\zeta(\omega_e) = \frac{S_\zeta(\omega)}{\left|\frac{d\omega_e}{d\omega}\right|}.$$

The transformed values are approximately:

ω [rad/s]	0.3	0.4	0.5	0.6
ω_e [rad/s]	0.347	0.484	0.631	0.789
$S^*_\zeta(\omega_e)$ [m^{2}s/rad]	0.380	1.409	0.492	0.153

Example: "crude hand calculations"

Transcribed task. Based on a wave spectrum and a heave transfer function, estimate:

1) the significant wave height,
2) the probability that the heave motion exceeds a level corresponding to three times the standard deviation of the heave motion.

Beam sea is assumed.

The slide estimates the wave-spectrum area as

$$m_0 \approx 0.6 \times 5.5 = 3.3\,\mathrm{m}^2,$$

so

$$H_s \approx 4\sqrt{3.3} = \boxed{7.3\,\mathrm{m}}.$$

For the response part, the hand calculation shown in the uploaded page builds a rough response spectrum $S_R(\omega) = |RAO|^2 S_\zeta(\omega)$ and estimates its area as approximately

$$m_{0R} \approx 1.68\,\mathrm{m}^2.$$

Thus the heave standard deviation is

$$\sigma_R = \sqrt{m_{0R}} = \sqrt{1.68} = 1.30\,\mathrm{m}.$$

If the exceedance level is $3\sigma_R$, then assuming Rayleigh-distributed peaks,

$$P(Z > 3\sigma_R) = \exp\left(-\frac{(3\sigma_R)^2}{2\sigma_R^2}\right) = e^{-9/2} = 0.011.$$

Therefore the exceedance probability is approximately

$$\boxed{1.1\%.}$$

MATLAB functionality

MatlabExample__VarianceSpectrum.m

This script is a compact teaching example for variance spectra and time-history reconstruction.

- It defines a set of regular-wave amplitudes `a`, frequencies `om`, and random phases `delta`.
- It computes the variance contribution of each component as

  $$\texttt{VarAmp} = \frac{1}{2}a^2,$$

  which is exactly the variance diagram idea from the lecture.
- It generates a time history by summing the individual cosine components.
- It then recomputes Fourier coefficients `A(n)` and `B(n)` numerically and reconstructs the signal as a Fourier approximation.
- Finally, it plots the variance diagram, several component time histories, and the full sea-surface elevation comparing direct simulation and Fourier reconstruction.

So the main teaching purpose of the code is to show the link between a discrete variance spectrum, a time-domain signal, and a Fourier-series representation.

NumIntegration.m

This script is the numerical-integration solution to the waterplane example.

- It stores tabulated breadth data.
- It uses `trapz` to evaluate the waterplane area.
- It computes the first moment of area and divides by the area to get the geometric centre / centre of flotation.
- It also shows that shifting the origin does not change the physical answer, only the coordinate value relative to the chosen reference.

StabilityExercise.m

This script solves the wall-sided barge problem from *Problem 1: Floatation and stability.*

- Question 1: draught from buoyancy balance.
- Question 2: ∇, KG, KB, waterplane inertia I, metacentric radius BMT, and metacentric height GMT.
- Question 3: restoring moment using the small-angle metacentric formula.
- Question 4: restoring moment using the wall-sided correction.

It is therefore a clean code implementation of the exact hand-solution sequence.

BendingMomentBarge.m (partial script from PDF)

This is the most important MATLAB workflow among the uploaded files because it links hydrostatics, spectra, transfer functions, and structural stress.

- It defines constants (ρ, g) and geometry (L, B, H, t).
- It computes the barge steel area and mass, then the total transported mass and draught.
- It sets the still-water vertical bending moment to zero because the total longitudinal load is uniform.
- It reads the RAO data from `VBM_RAO.xls`.
- It constructs the Bretschneider wave spectrum using H_s and T_z.
- It multiplies the wave spectrum by the squared RAO to form the response spectrum of bending moment.
- It integrates the response spectrum to obtain spectral moments, the zero-upcrossing period of the response, and the most probable largest bending moment over the specified storm duration.
- Finally, it computes the neutral axis, second moment of area, and deck/keel stresses.

The missing lines in the partial PDF are not a problem for understanding the workflow. The logic of the script is fully recoverable even if the external Excel file is not available.

Short list of corrected points and ambiguities

- In *Example: Probability of exceedance*, the handwritten 63% for crest exceedance is inconsistent with the stated variance; the correct value is 48.6%.
- In *Example: Visual/crude estimation....*, the wording says 3 hours but the actual computation uses 3600 s, i.e. 1 hour.
- In *Example: Transportation of a semi-submersible*, the RAO-based spectrum plot cannot be reproduced exactly without the external file `VBM_RAO.xls`. The uploaded partial script does, however, report the final most probable largest bending moment and the final stress levels.